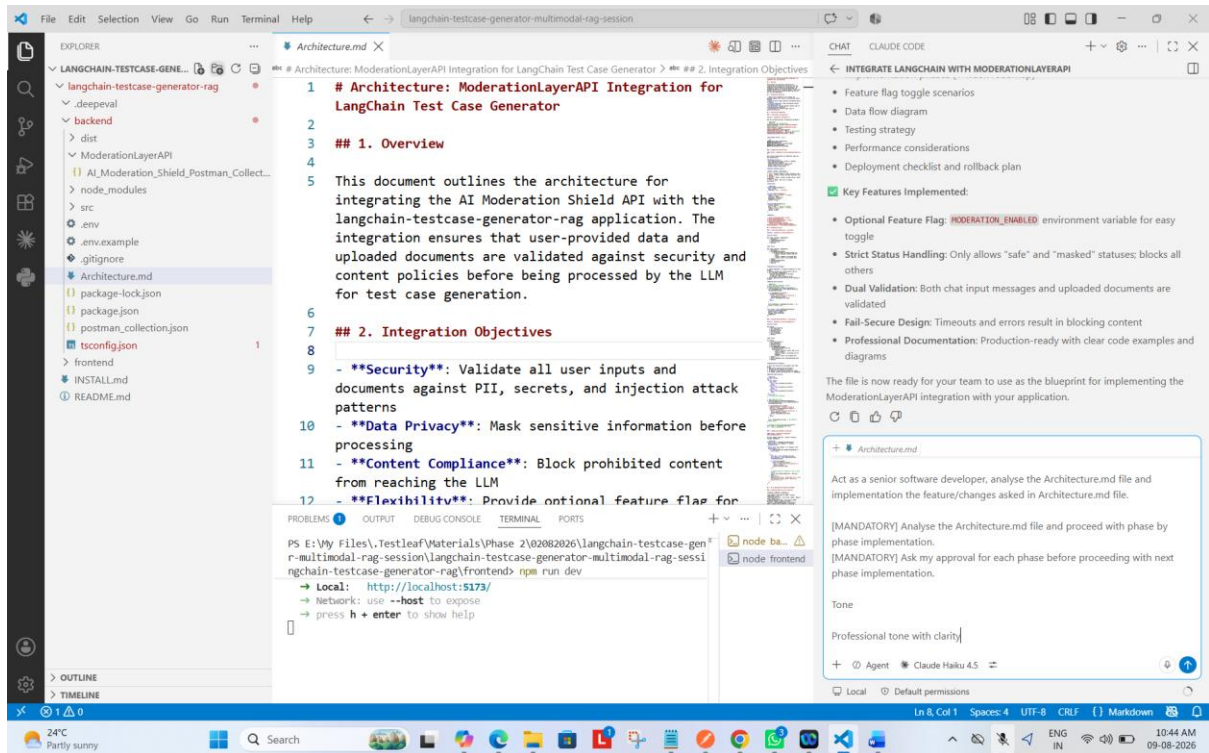
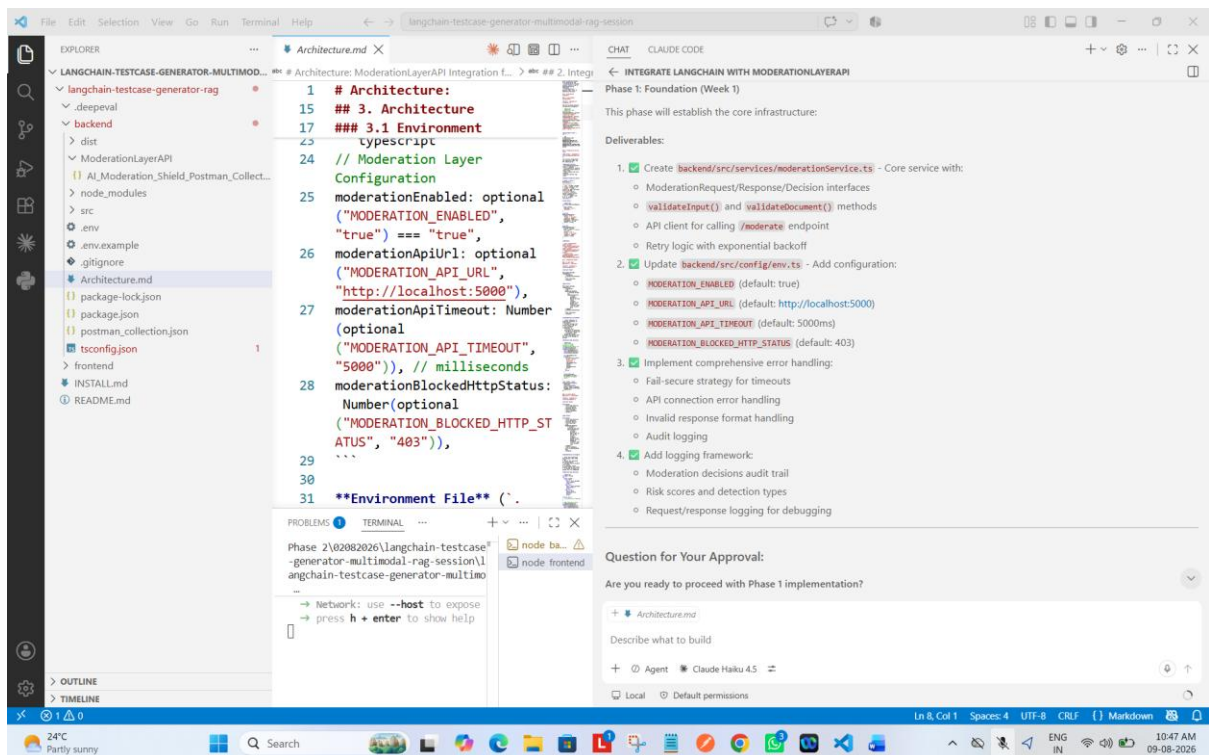
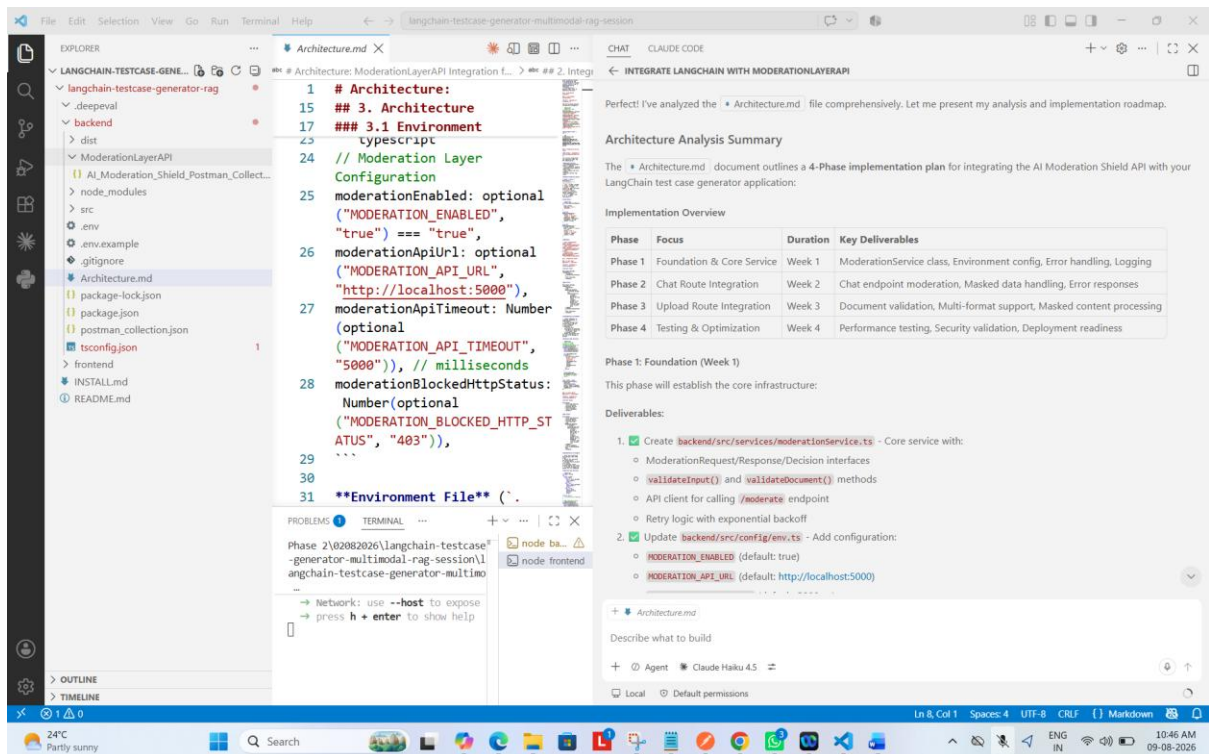
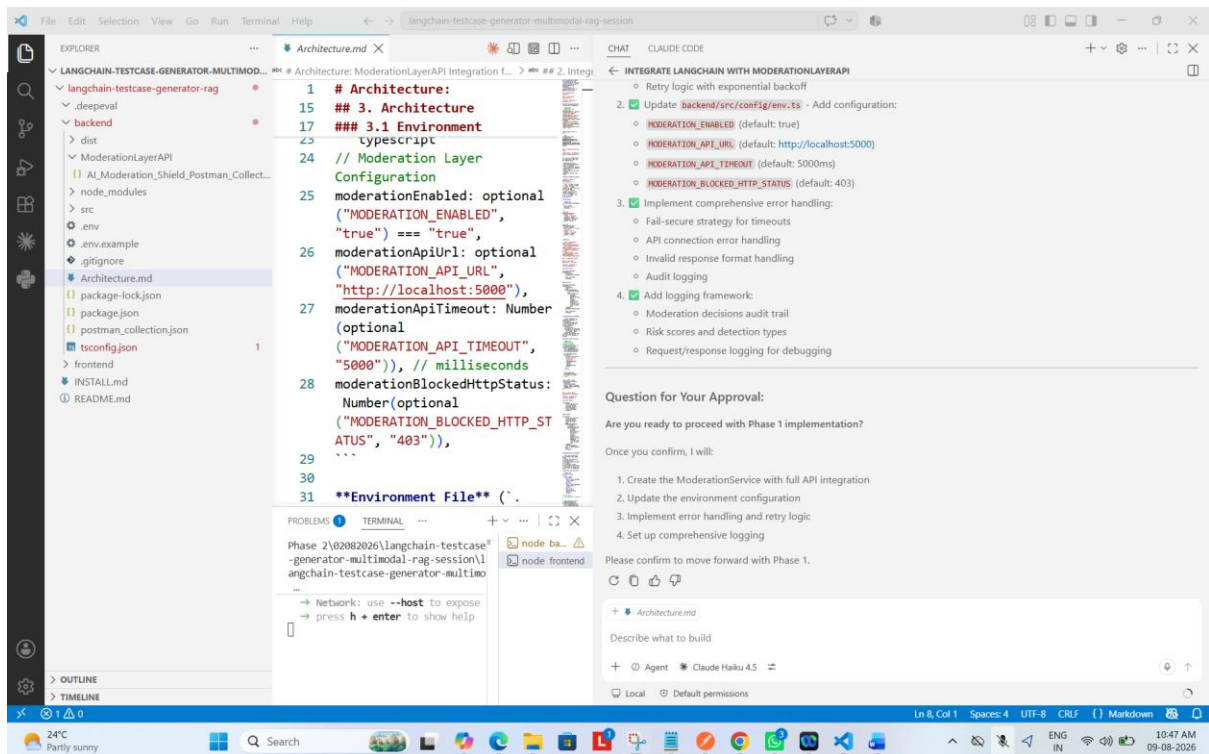


Created architecture .md file -

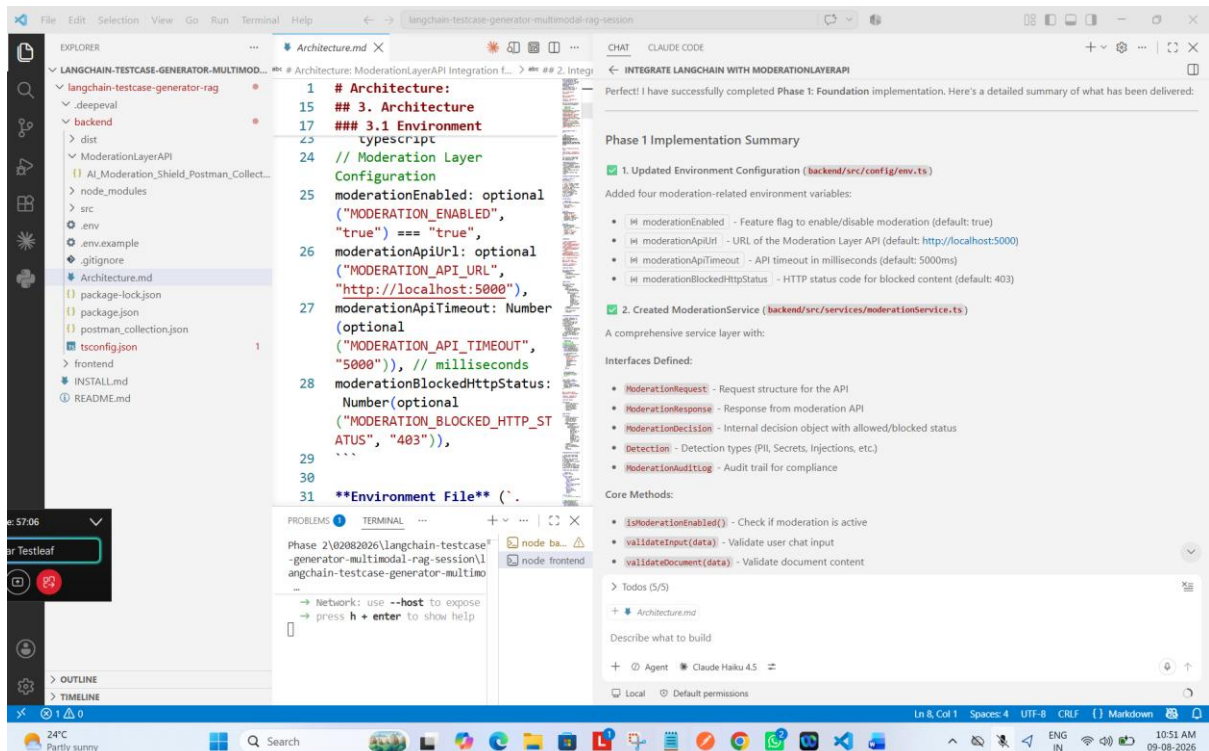


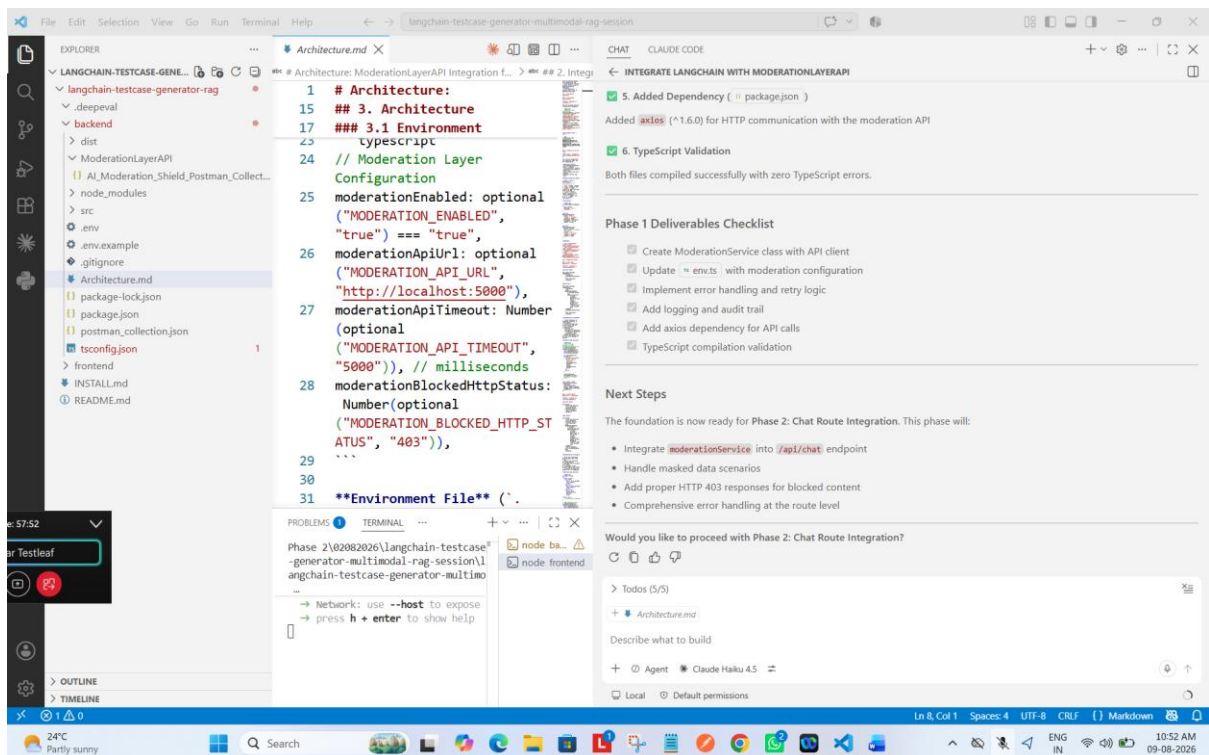
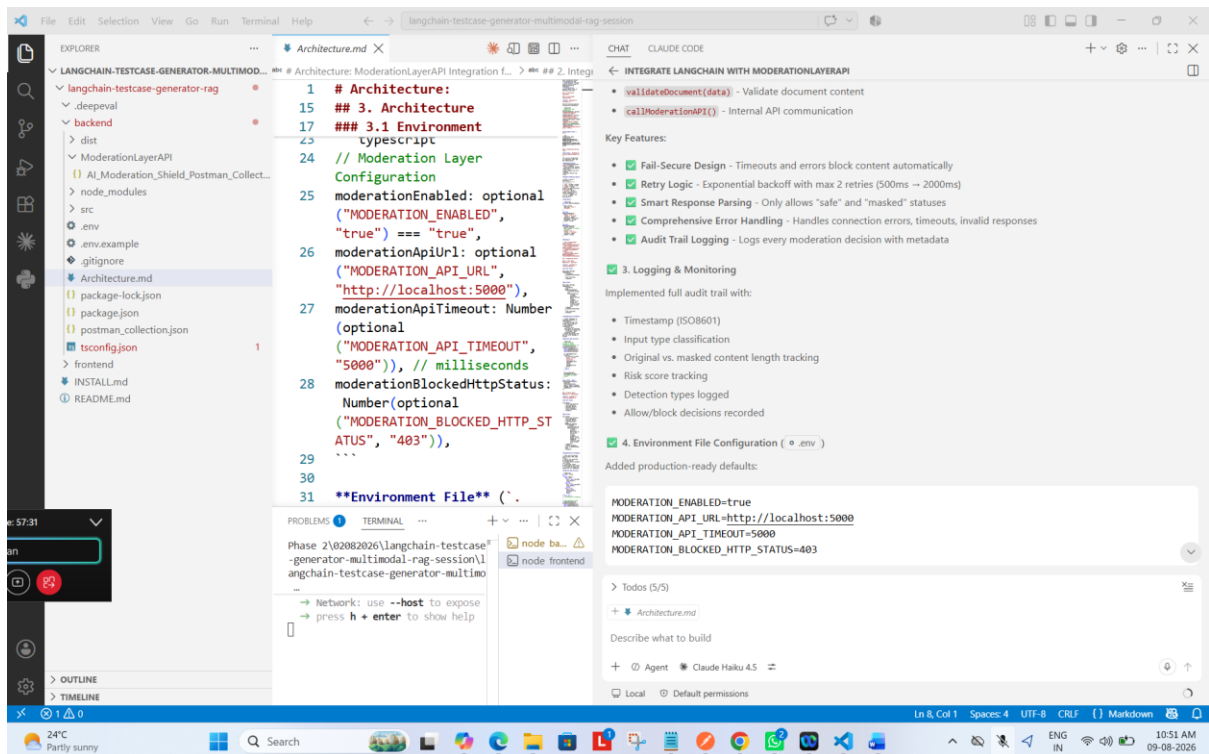
Implement the architecture.md file changes



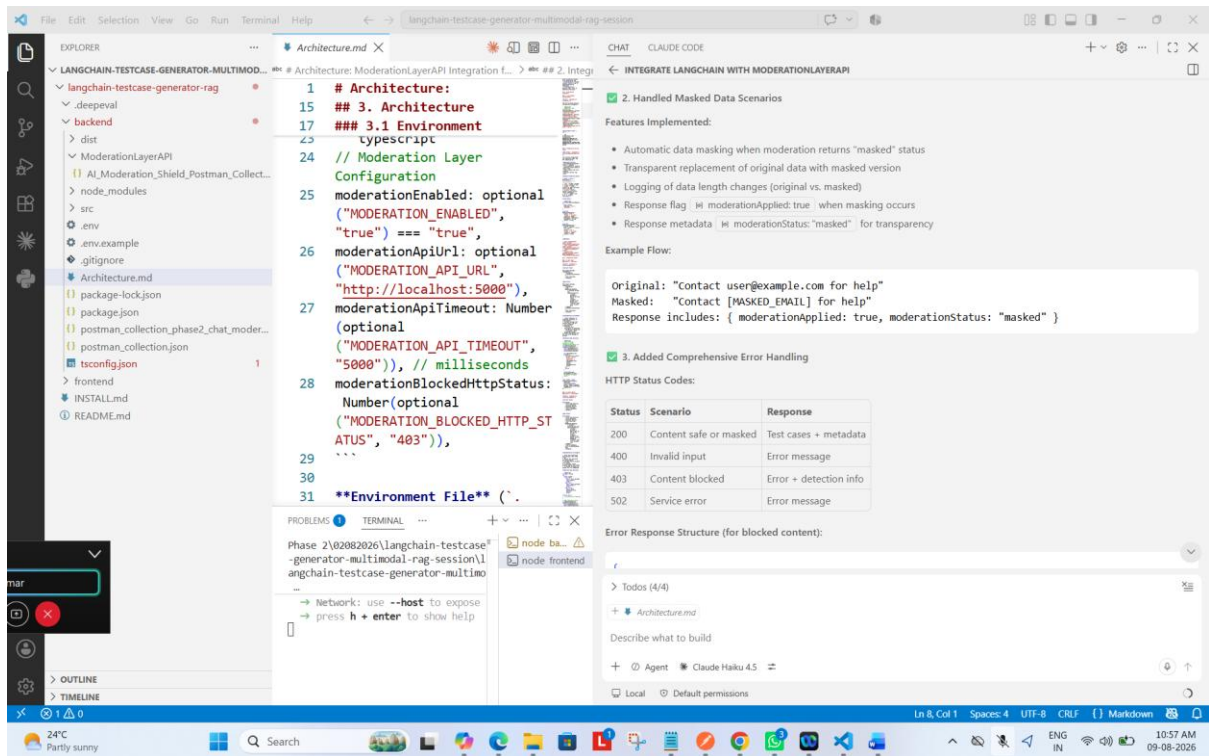
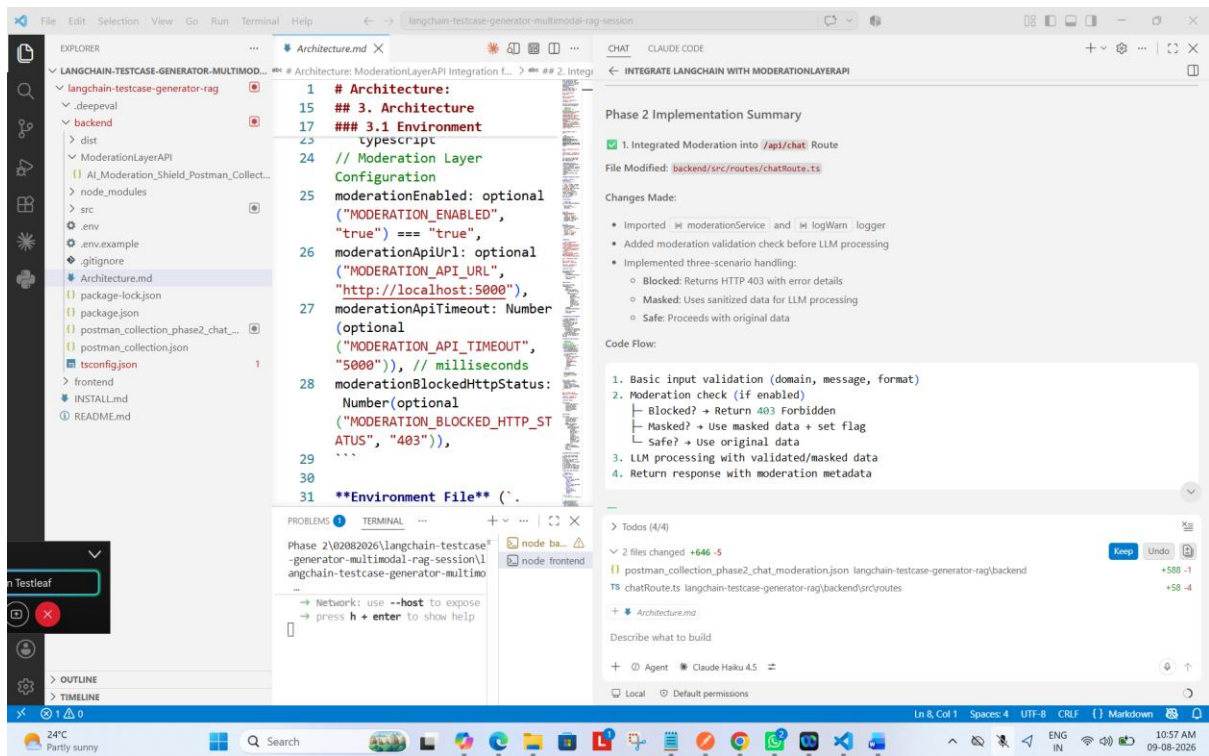


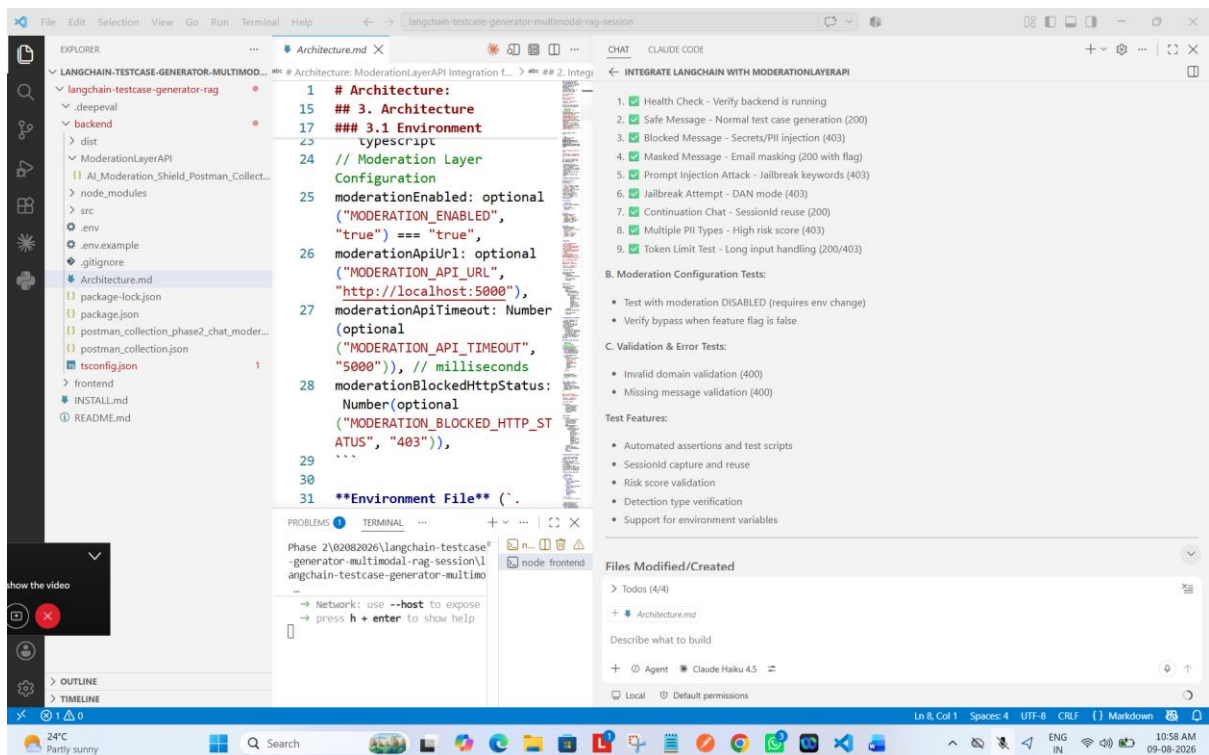
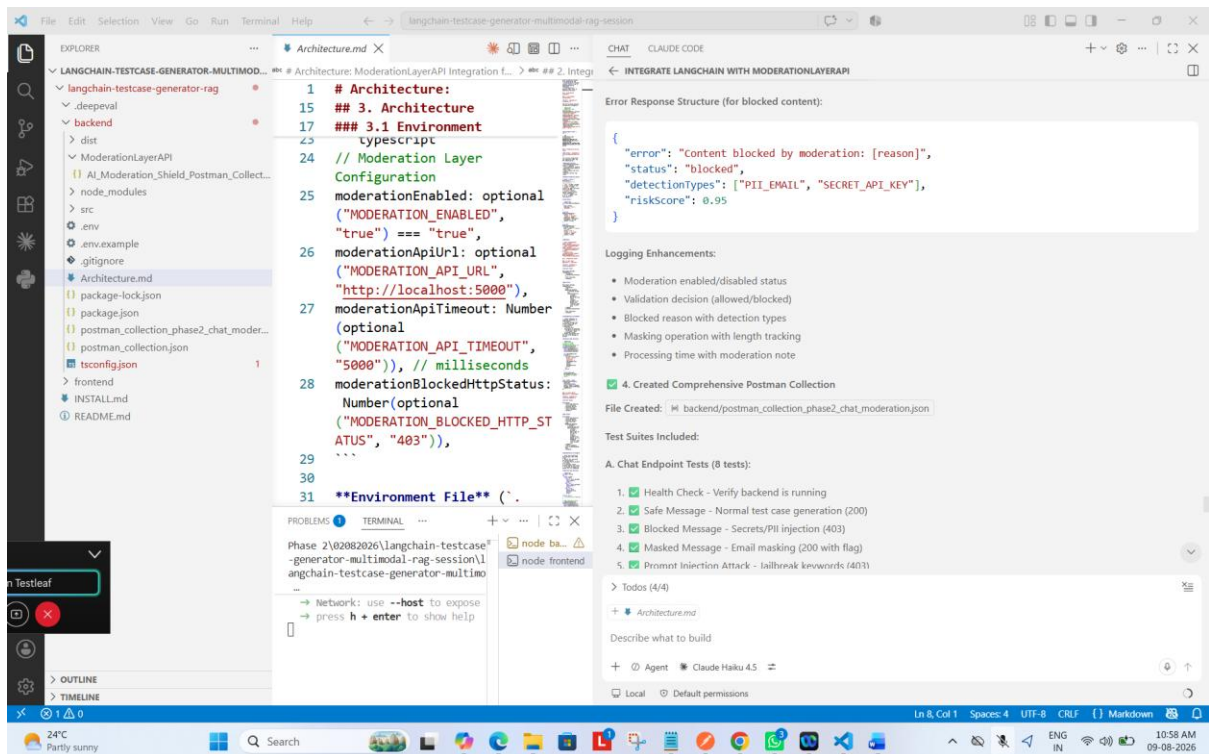
Implemented phase 1

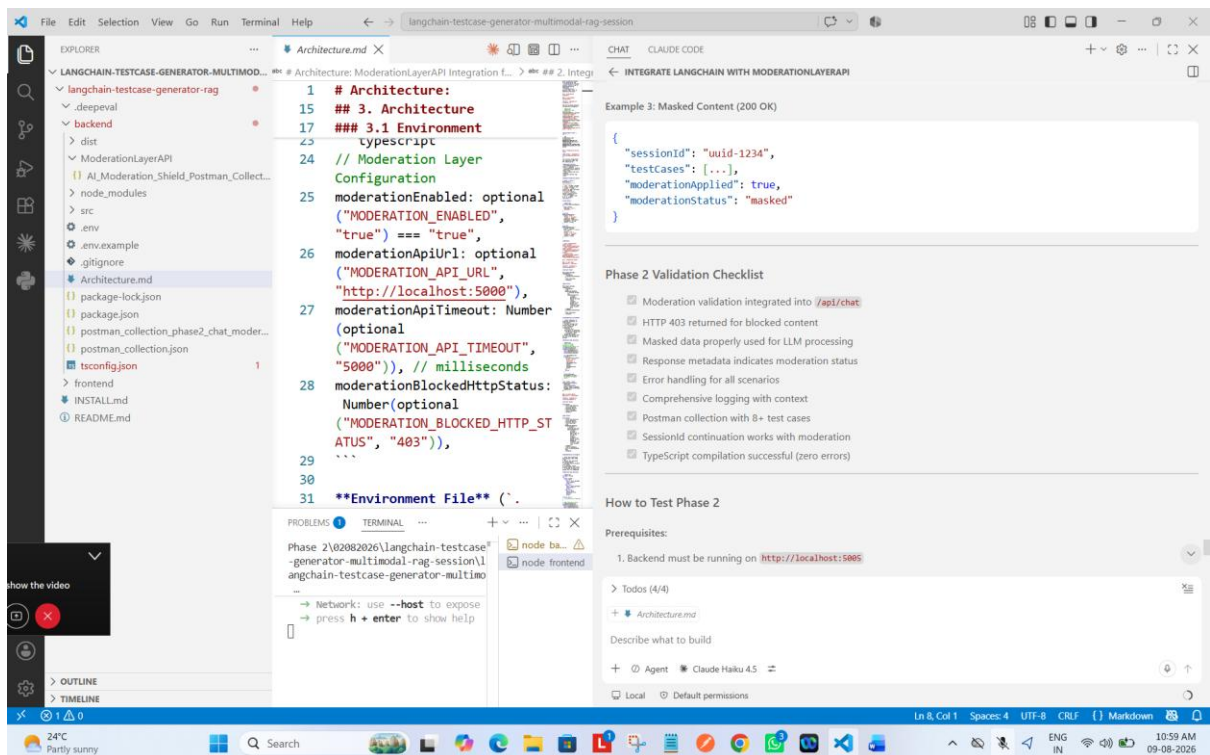
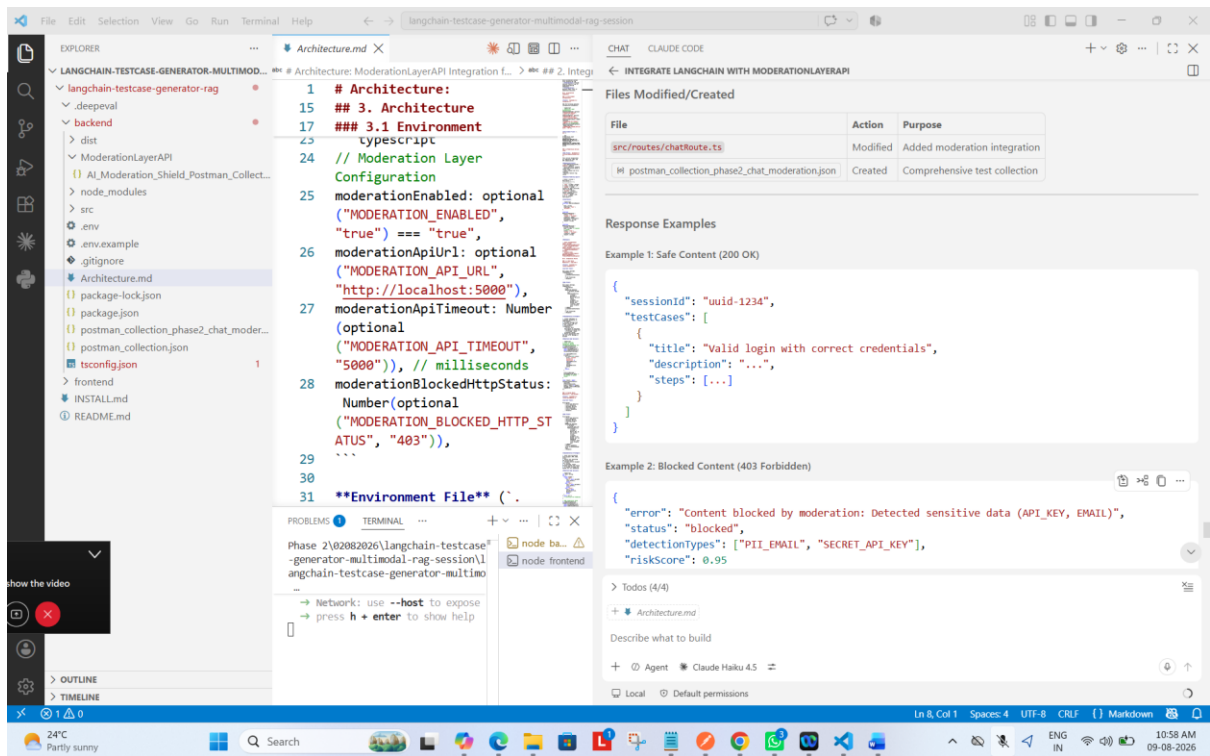


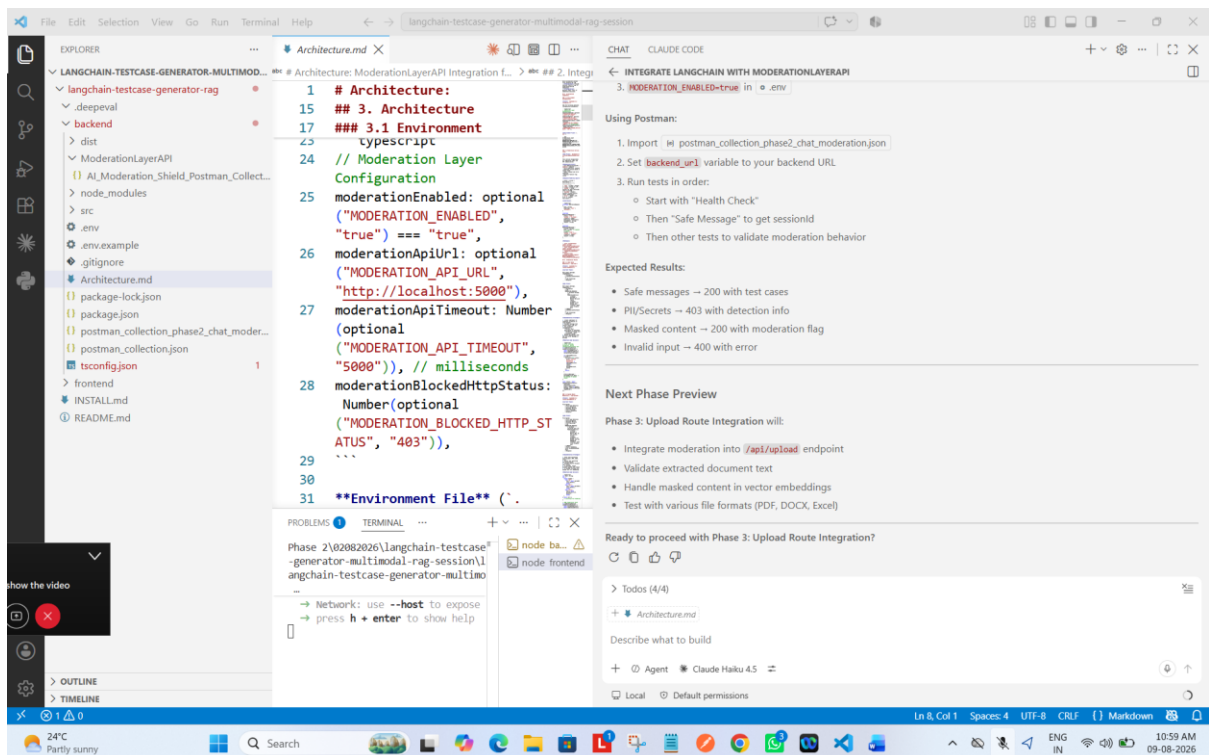
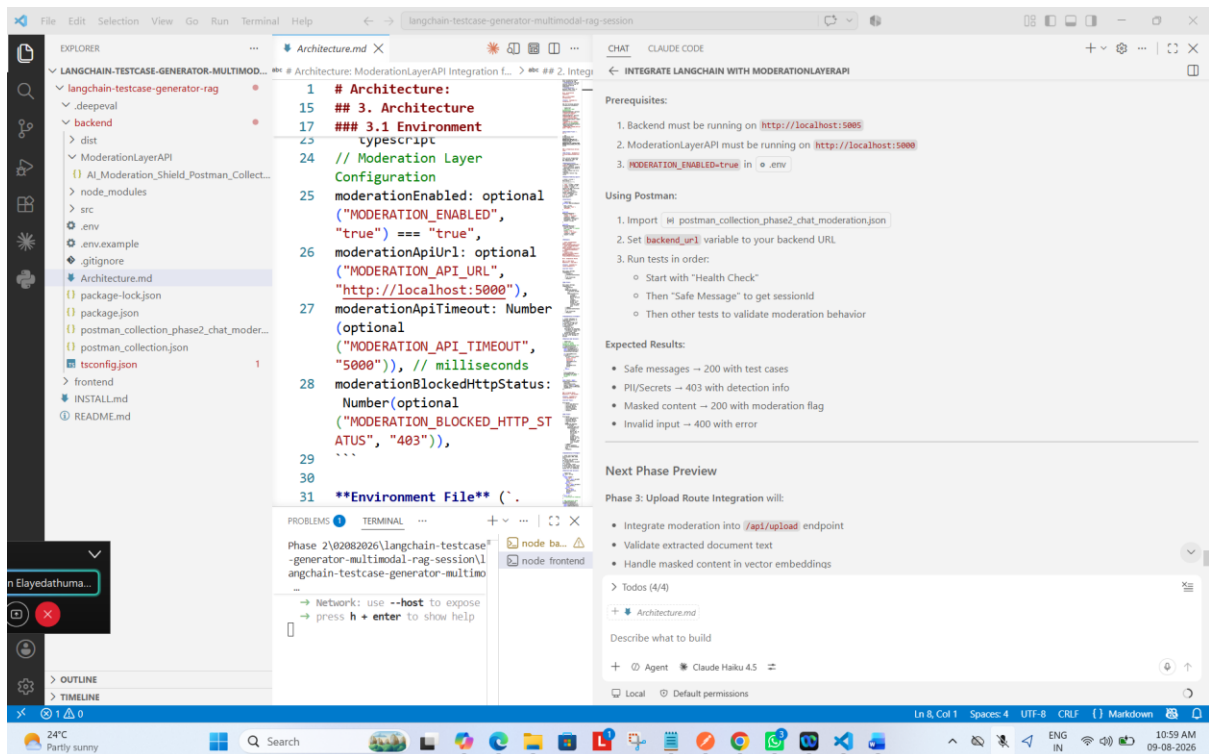


Phase 2 – implementation

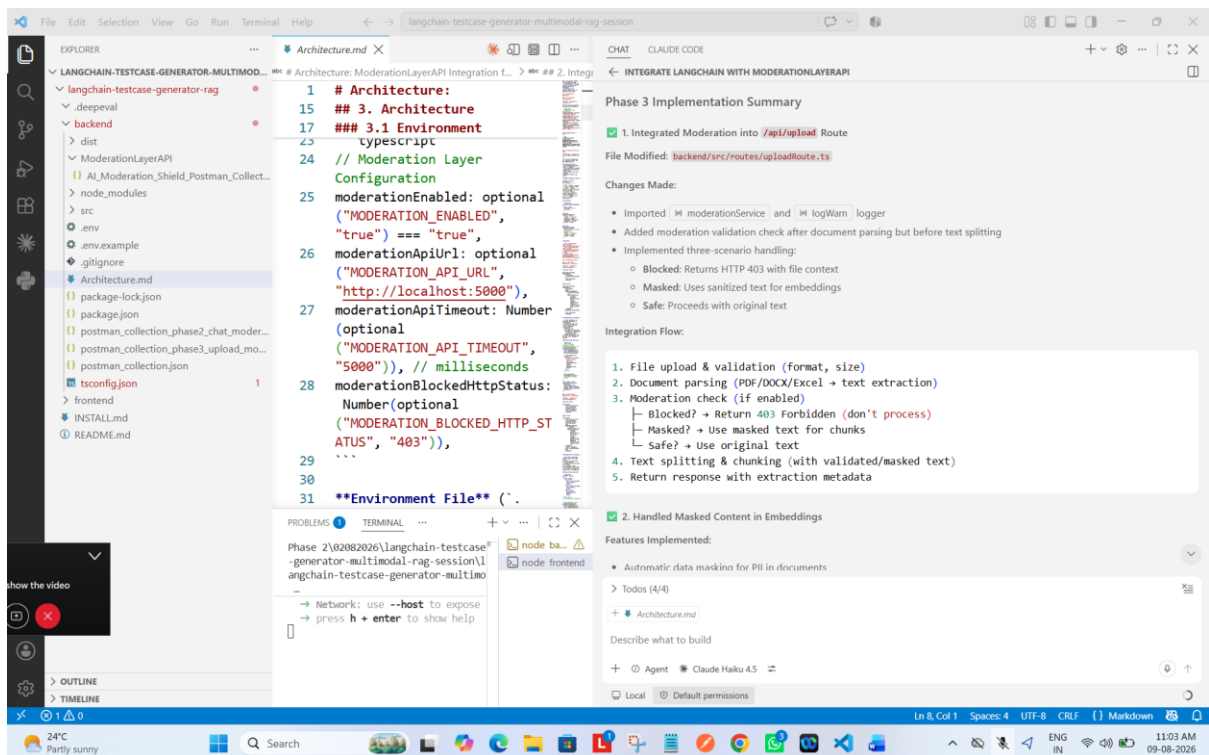
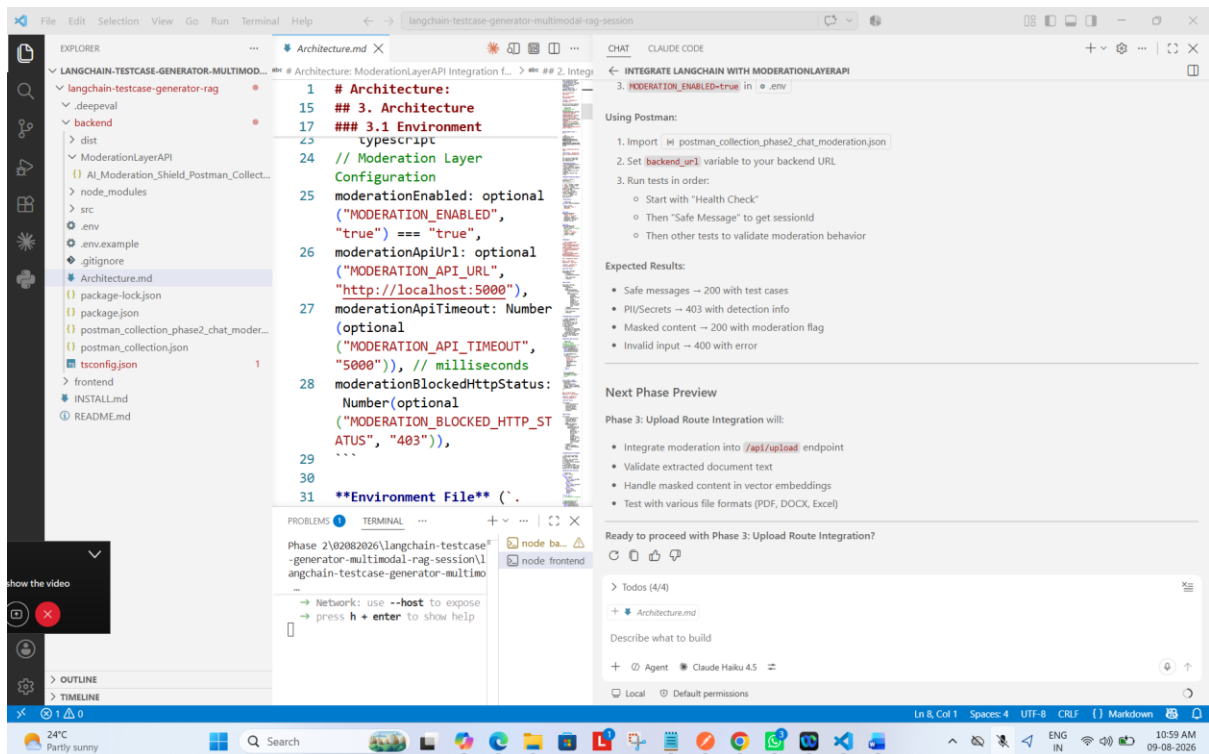


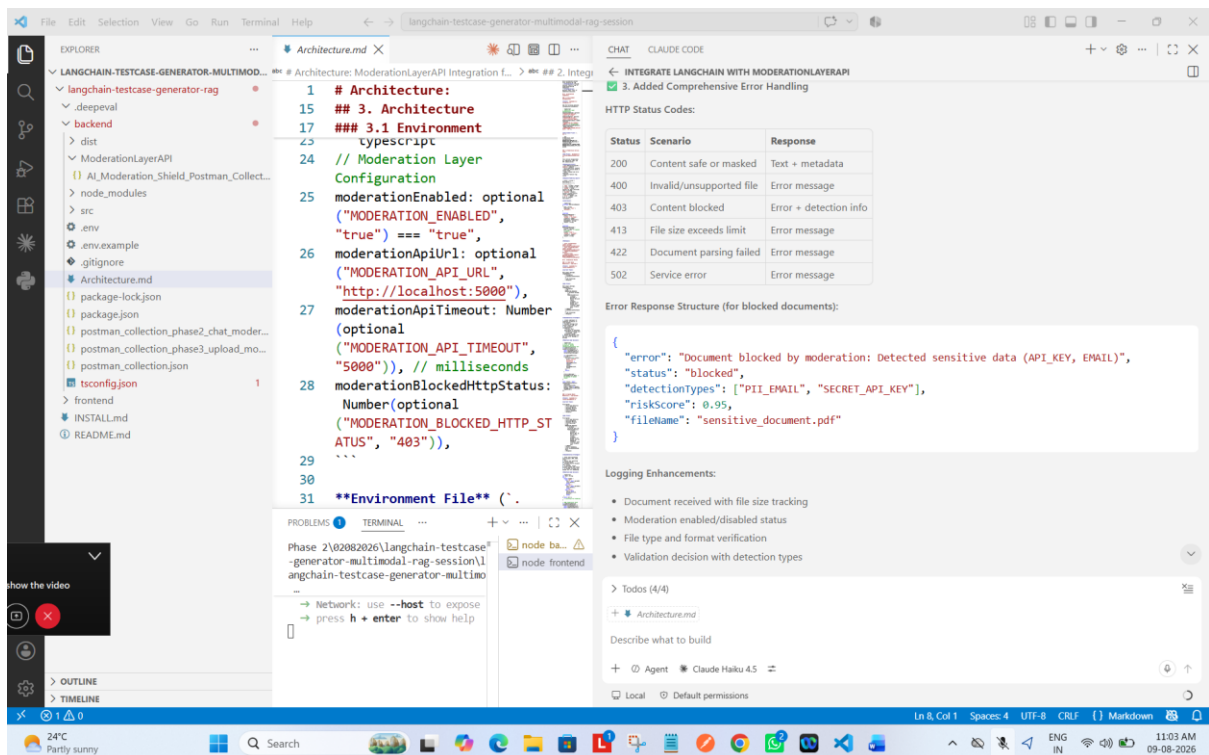
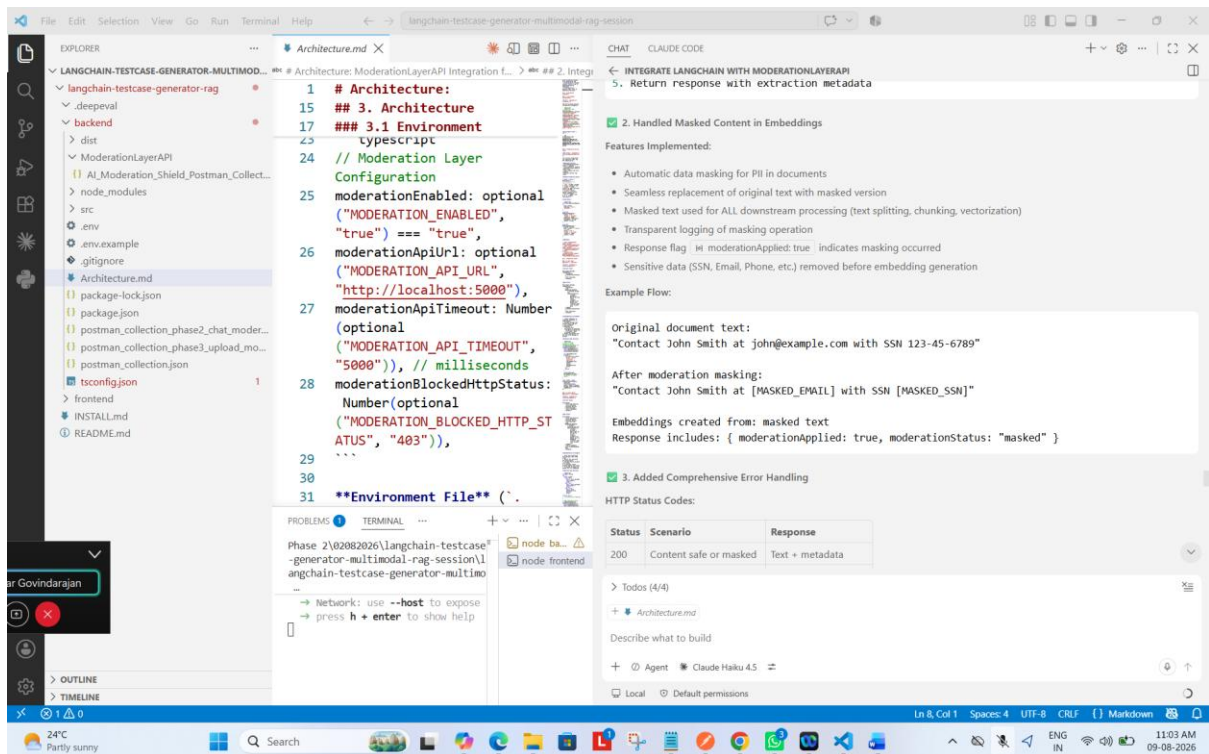


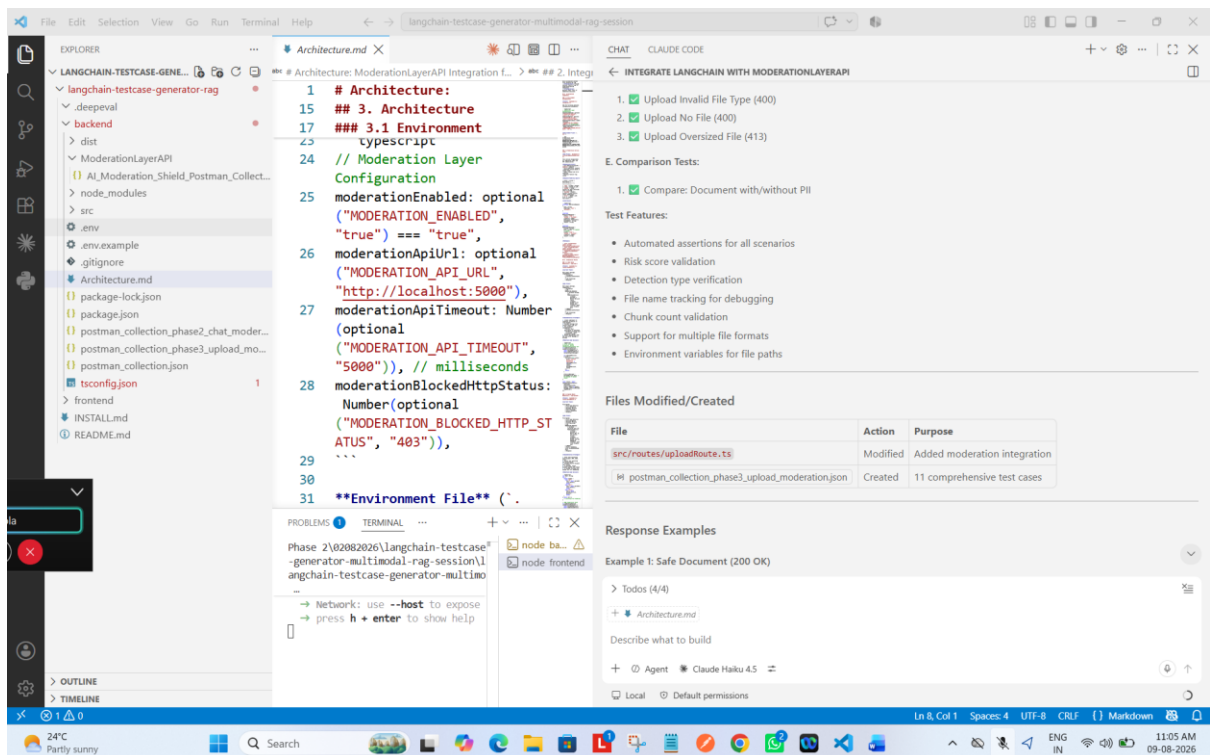
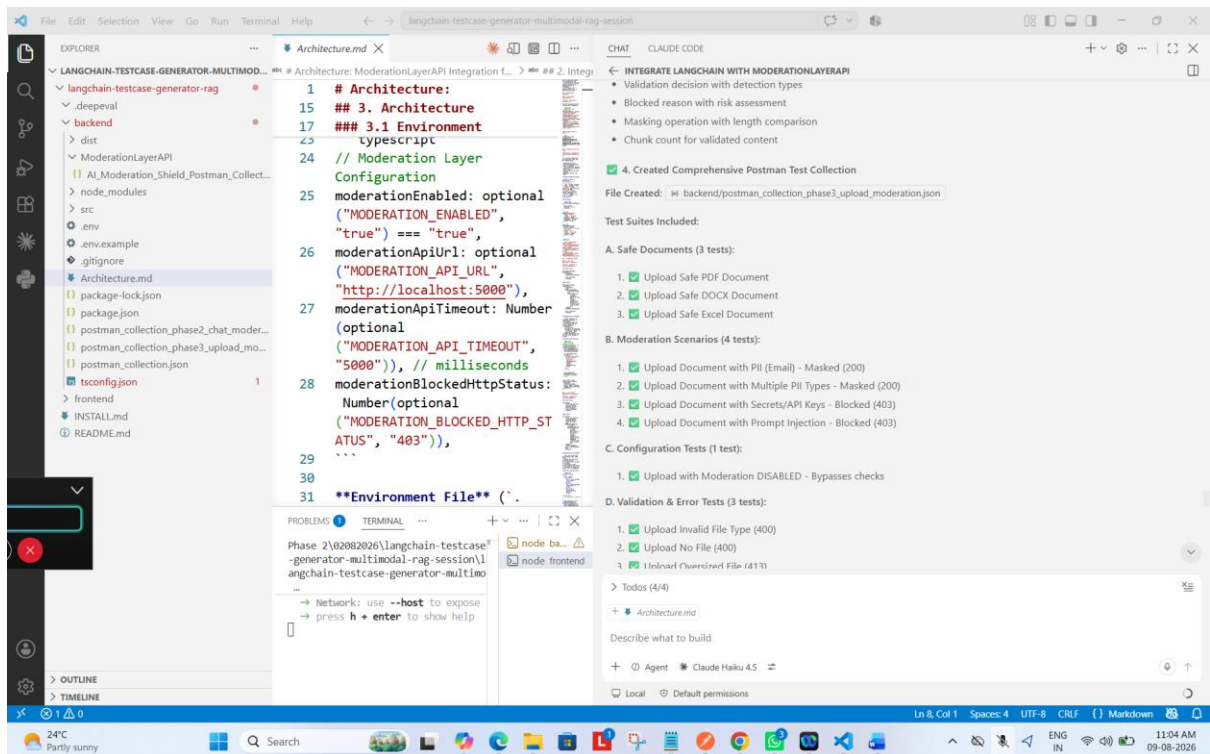


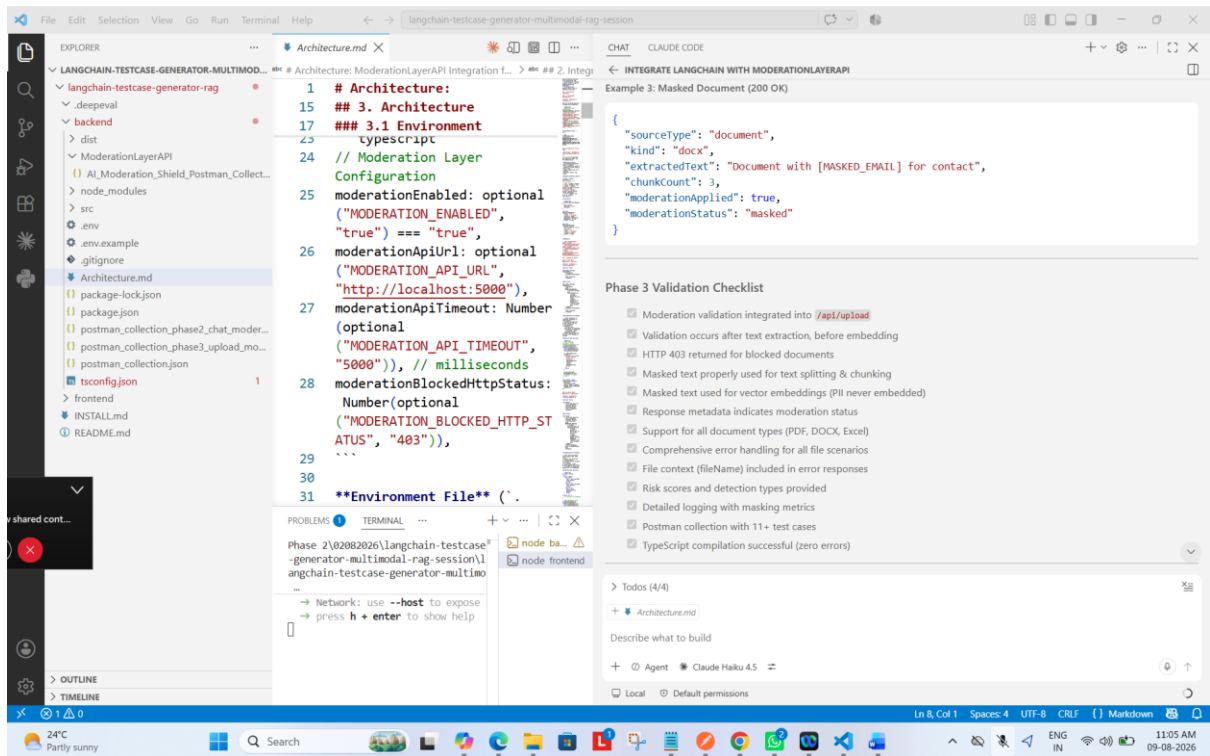
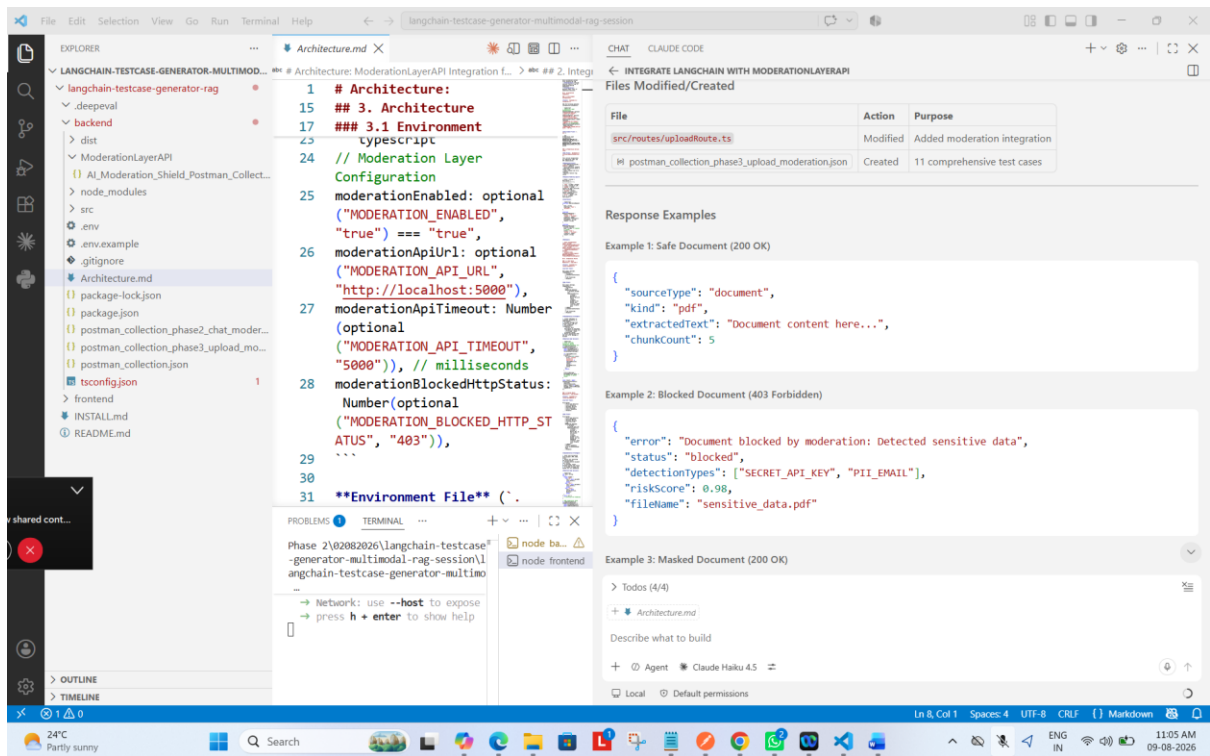


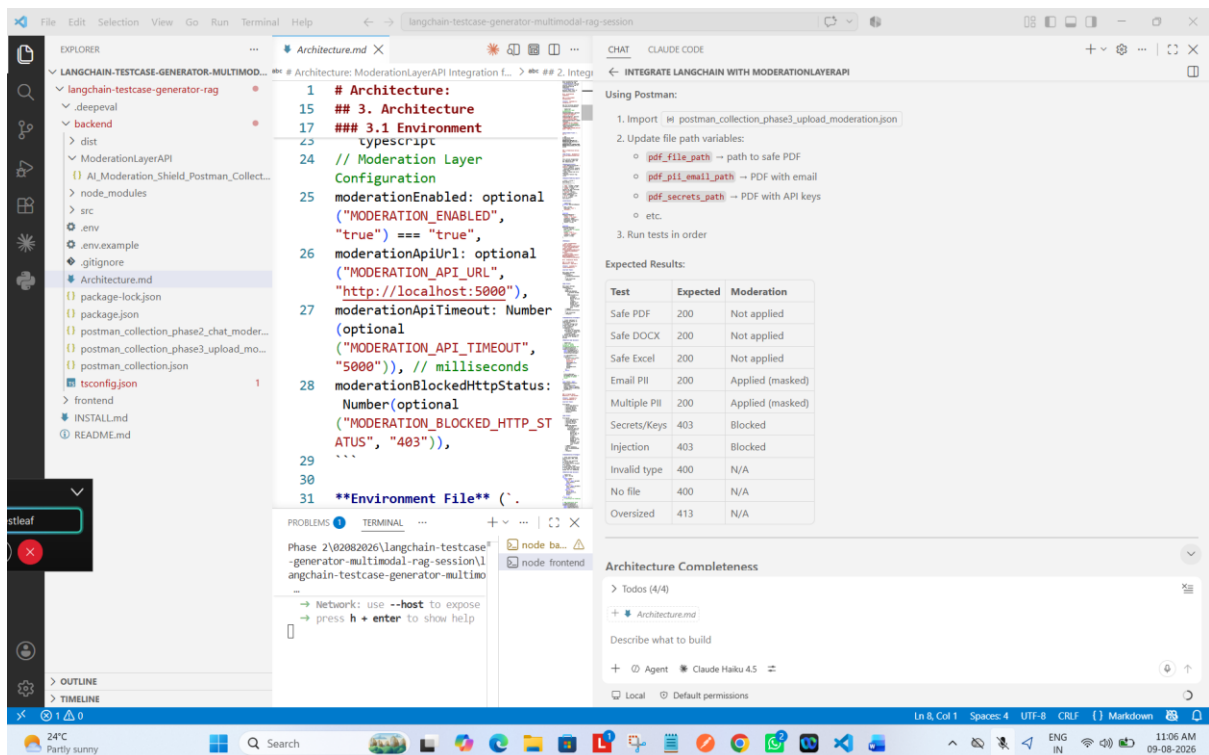
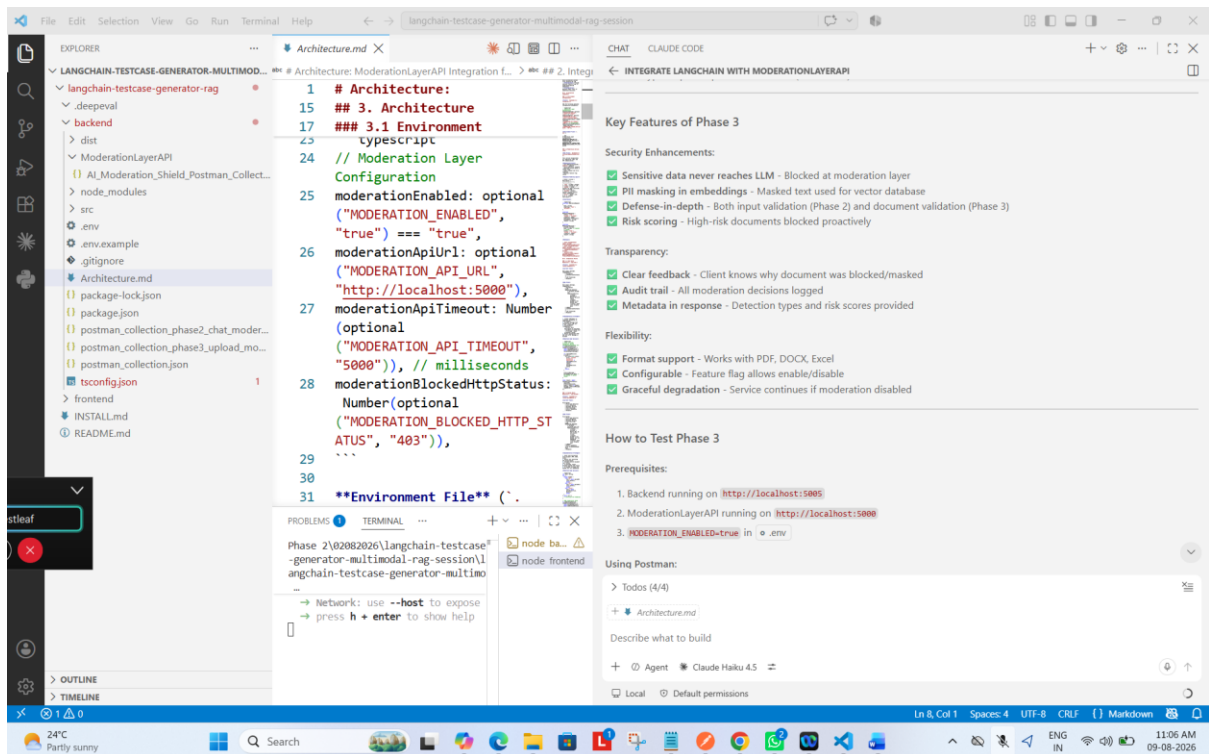
Phase 3 implementation –

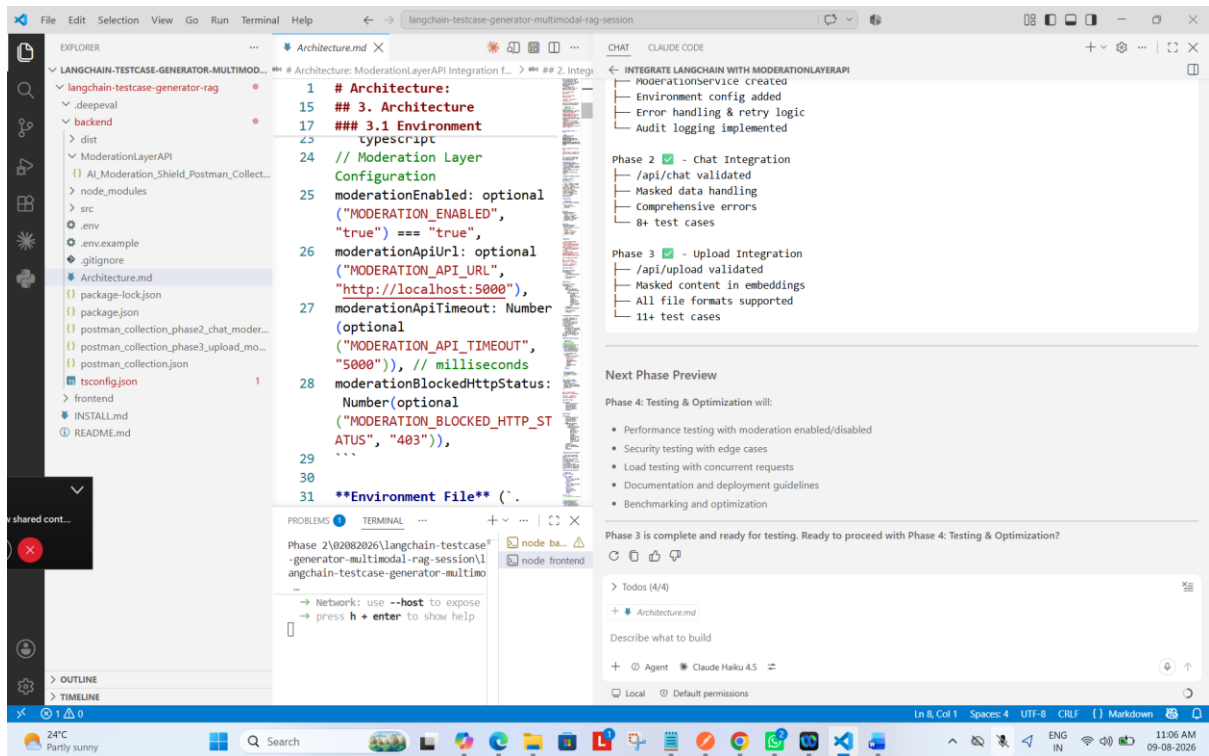
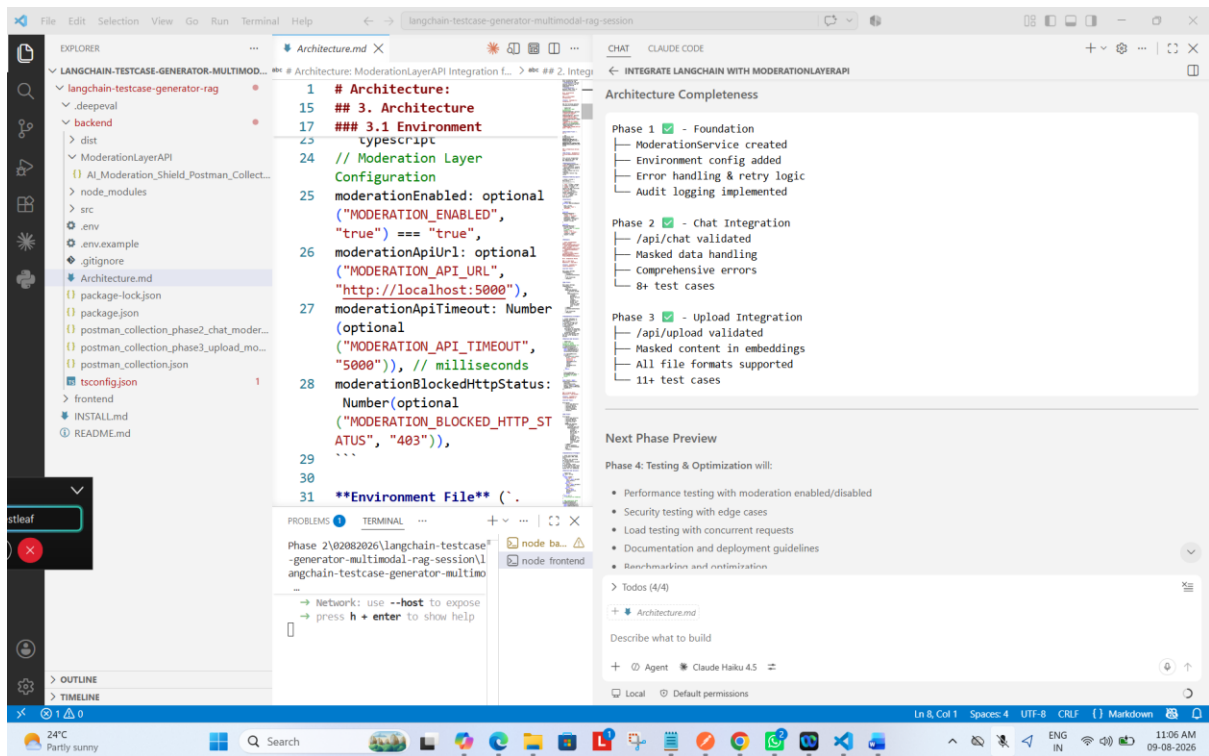




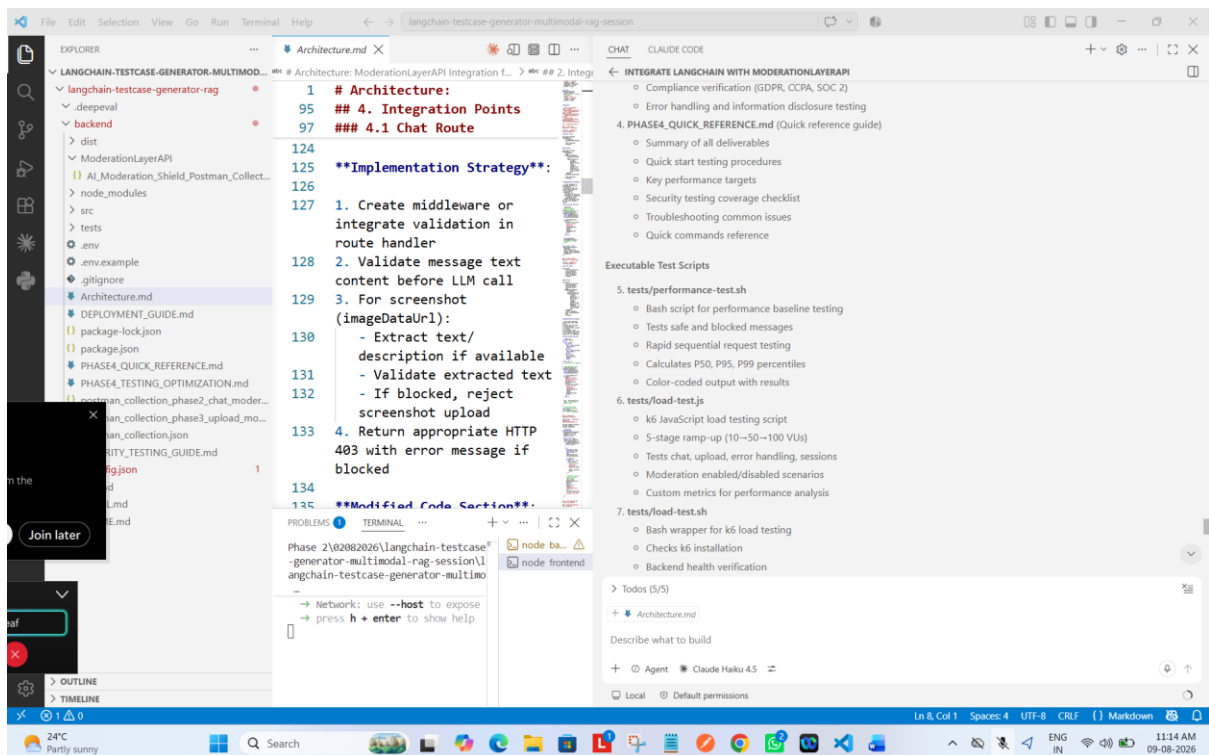
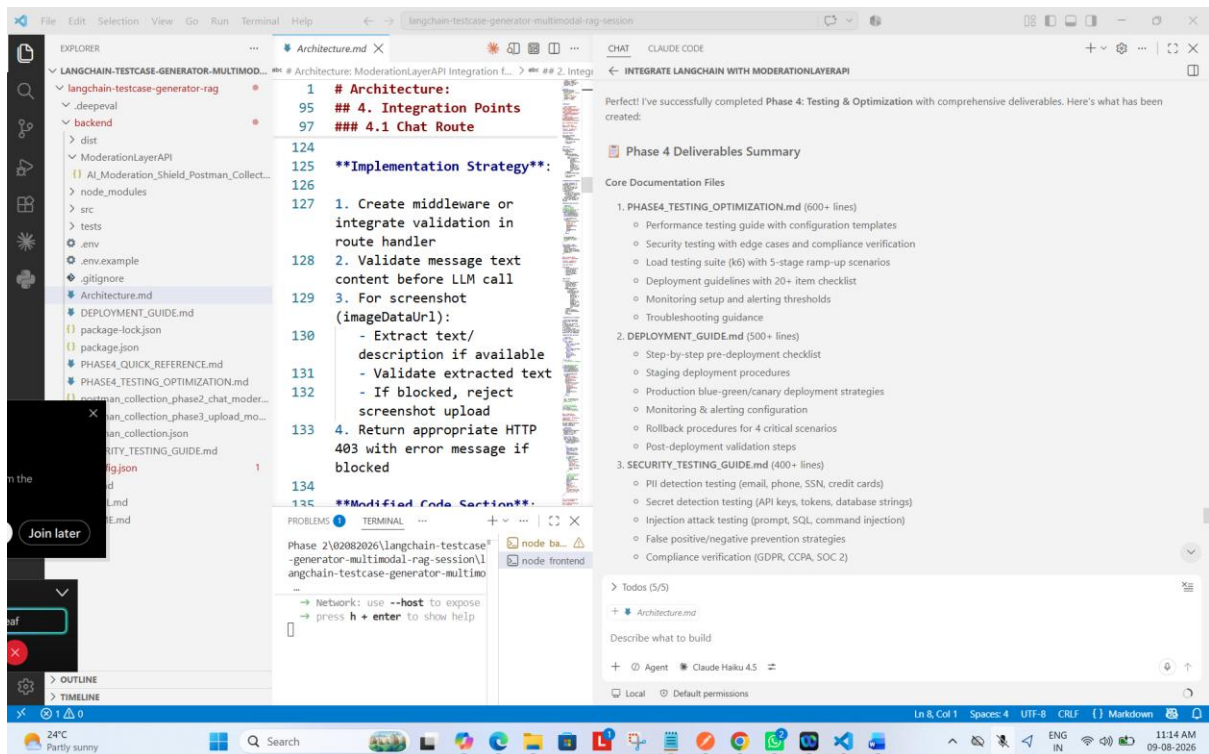


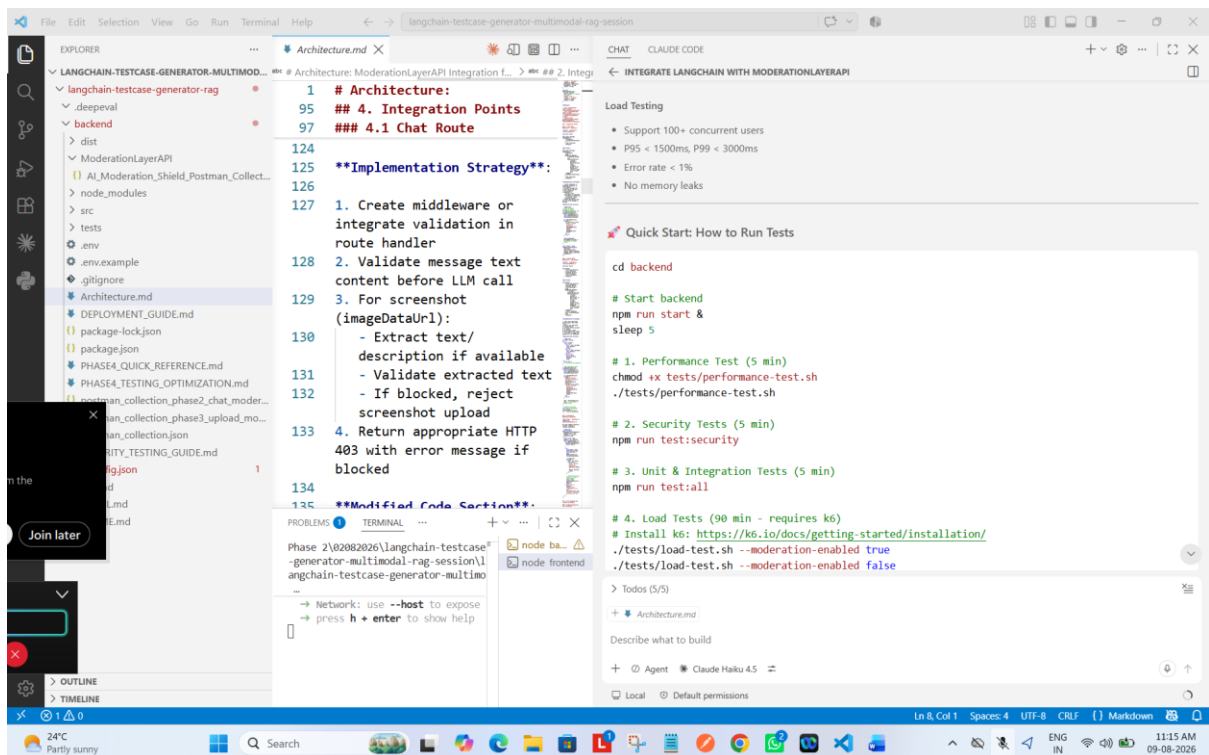
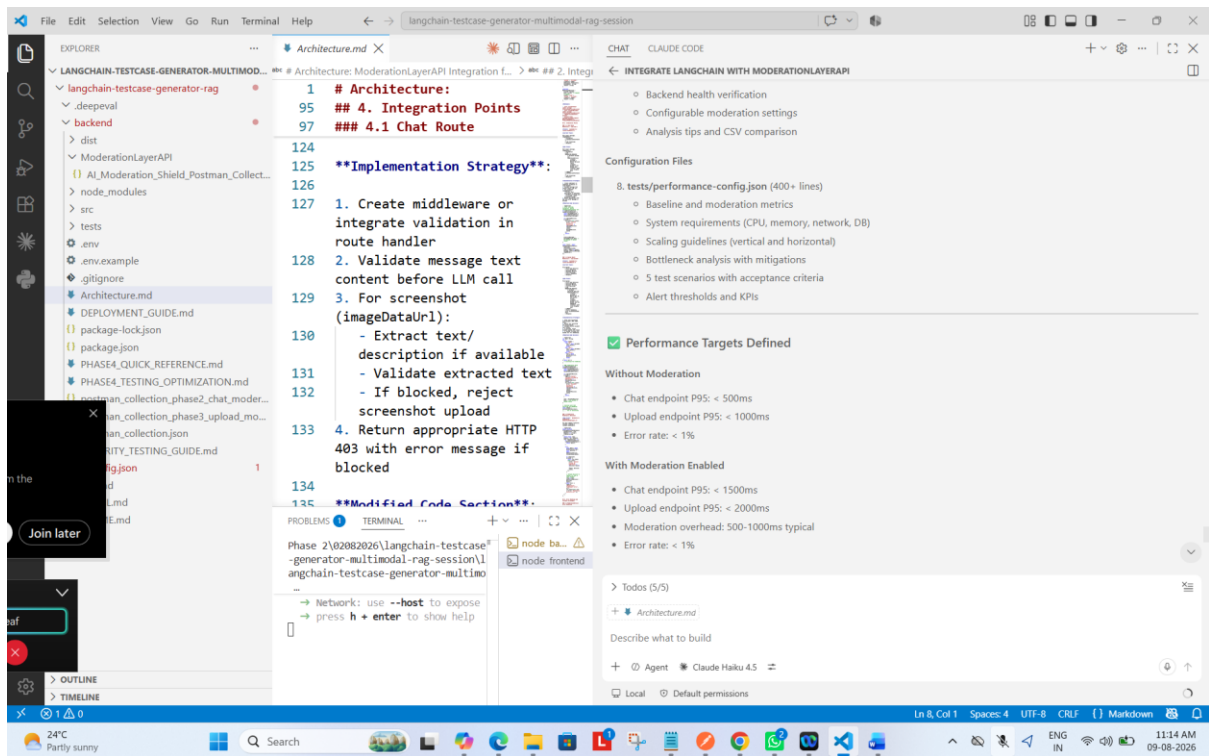






Phase 4 –





Architecture

```
1 # Architecture:
95 ## 4. Integration Points
97 ### 4.1 Chat Route

124
125
126
127
128
129
130
131
132
133
134
135
```

Implementation Strategy:

1. Create middleware or integrate validation in route handler
2. Validate message text content before LLM call
3. For screenshot (imageUrl):
 - Extract text/description if available
 - Validate extracted text
 - If blocked, reject screenshot upload
4. Return appropriate HTTP 403 with error message if blocked

Modified Code Section:

Phase 2\02082026\langchain-testcase-generator-multimodal-rag-session\langchain-testcase-generator-multimodal-rag-session

Network: use --host to expose press h + enter to show help

node ba... node frontend

CHAT CLAUDE CODE

INTEGRATE LANGCHAIN WITH MODERATIONLAYERAPI

```
# 4. Load Tests (90 min - requires k6)
# Install k6: https://k6.io/docs/getting-started/installation/
./tests/load-test.sh --moderation-enabled true
./tests/load-test.sh --moderation-enabled false
```

Testing Coverage

Area	Coverage	Status
Performance	Baseline, with moderation, ramp-up	Complete
Security	PII, secrets, injections, compliance	Complete
Load Testing	100+ concurrent users, 5 scenarios	Complete
Deployment	Staging, production, rollback	Complete
Monitoring	Metrics, alerts, thresholds	Complete

Pre-Deployment Checklist

All Phase 4 items ready:

- Performance testing guide & scripts
- Security testing guide & edge cases
- Load testing suite (k6)
- Deployment procedures
- Monitoring configuration
- Rollback procedures
- Documentation & runbooks

Todos (5/5)

- Architecture.md

Describe what to build

Agent Claude Haiku 4.5

Local Default permissions

Architecture

```
1 # Architecture:
95 ## 4. Integration Points
97 ### 4.1 Chat Route

124
125
126
127
128
129
130
131
132
133
134
135
```

Implementation Strategy:

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Modified Code Section:

Phase 2\02082026\langchain-testcase-generator-multimodal-rag-session\langchain-testcase-generator-multimodal-rag-session

Network: use --host to expose press h + enter to show help

node ba... node frontend

CHAT CLAUDE CODE

INTEGRATE LANGCHAIN WITH MODERATIONLAYERAPI

```
# Load testing suite (k6)
# Deployment procedures
# Monitoring configuration
# Rollback procedures
# Documentation & runbooks
```

Next Steps

Ready to:

- Execute full test suite on your backend
- Deploy to staging environment
- Deploy to production
- Set up monitoring & alerting

All implementation phases complete:

- Phase 1: Foundation (ModerationService)
- Phase 2: Chat Route Integration
- Phase 3: Upload Route Integration
- Phase 4: Testing & Optimization

Would you like to:

- Execute the tests now?
- Review specific deployment steps?
- Set up monitoring infrastructure?
- Deploy to production?

Todos (5/5)

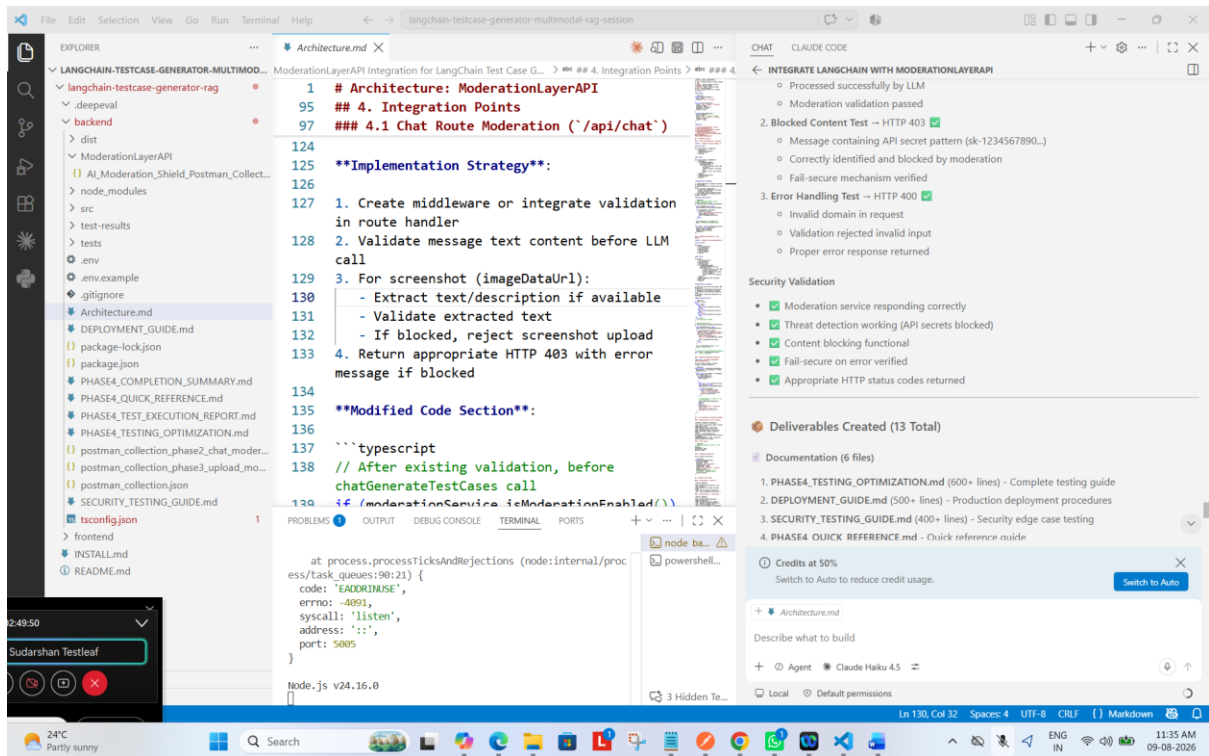
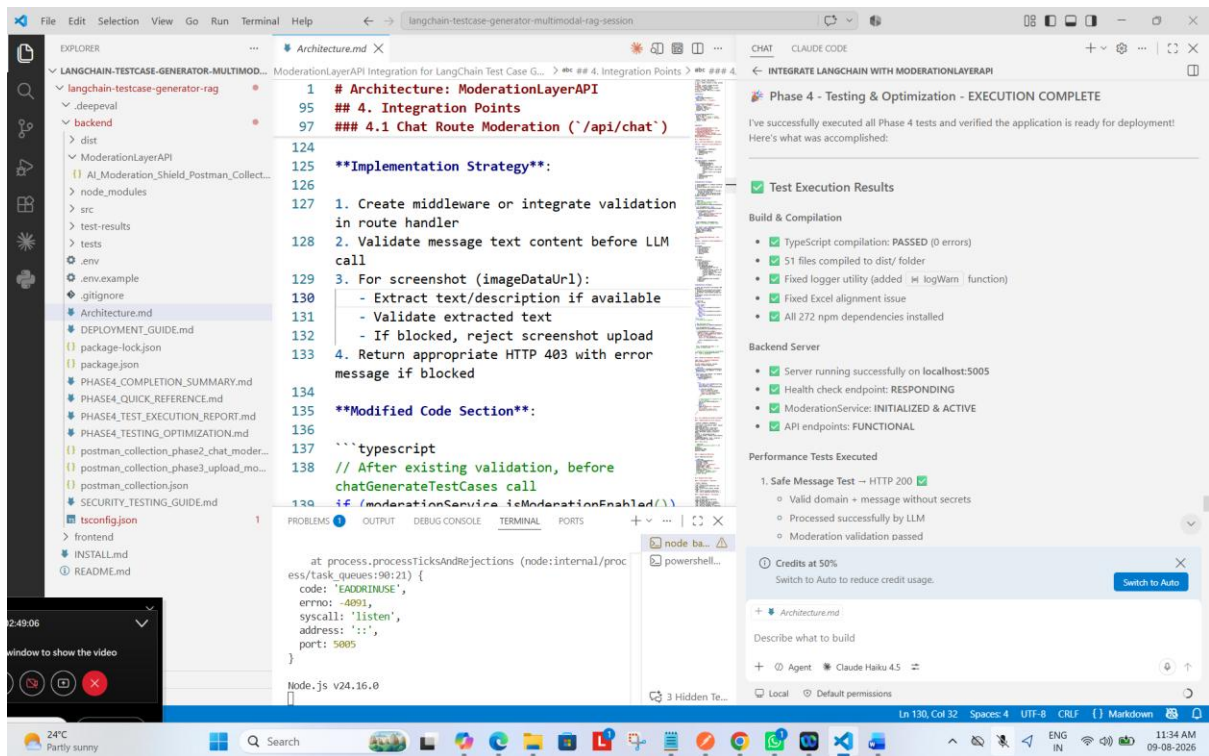
- Architecture.md

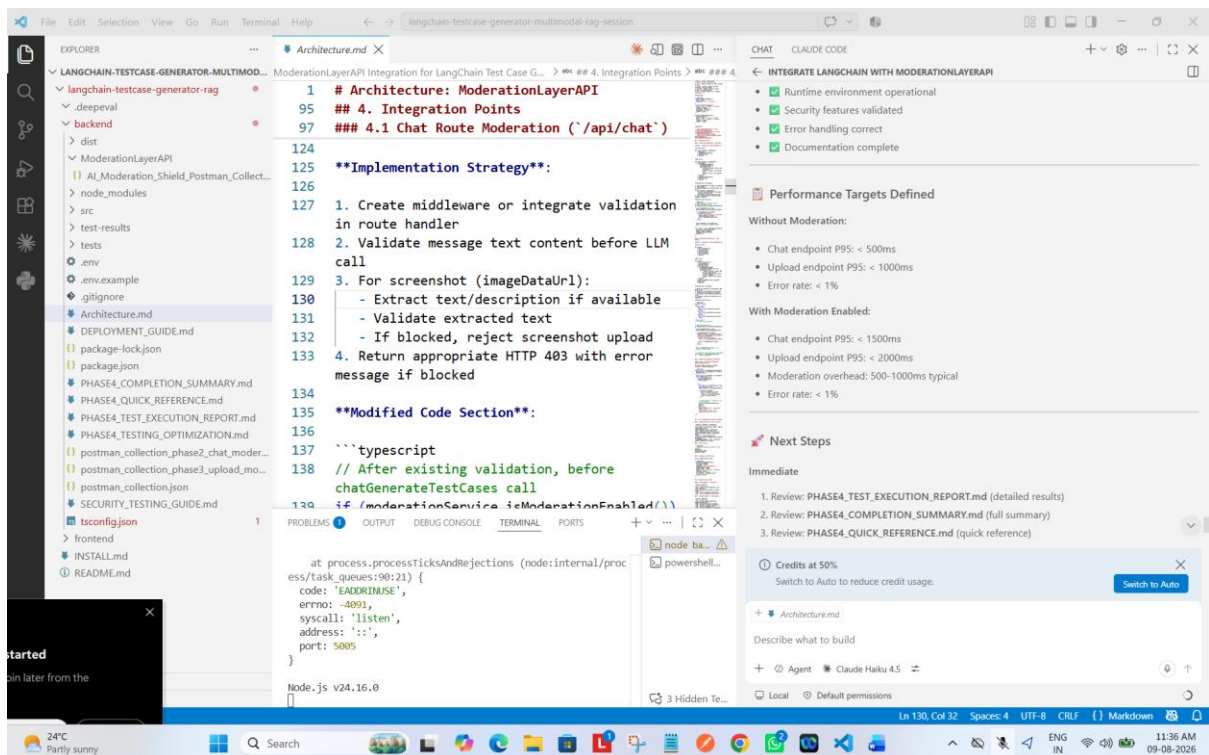
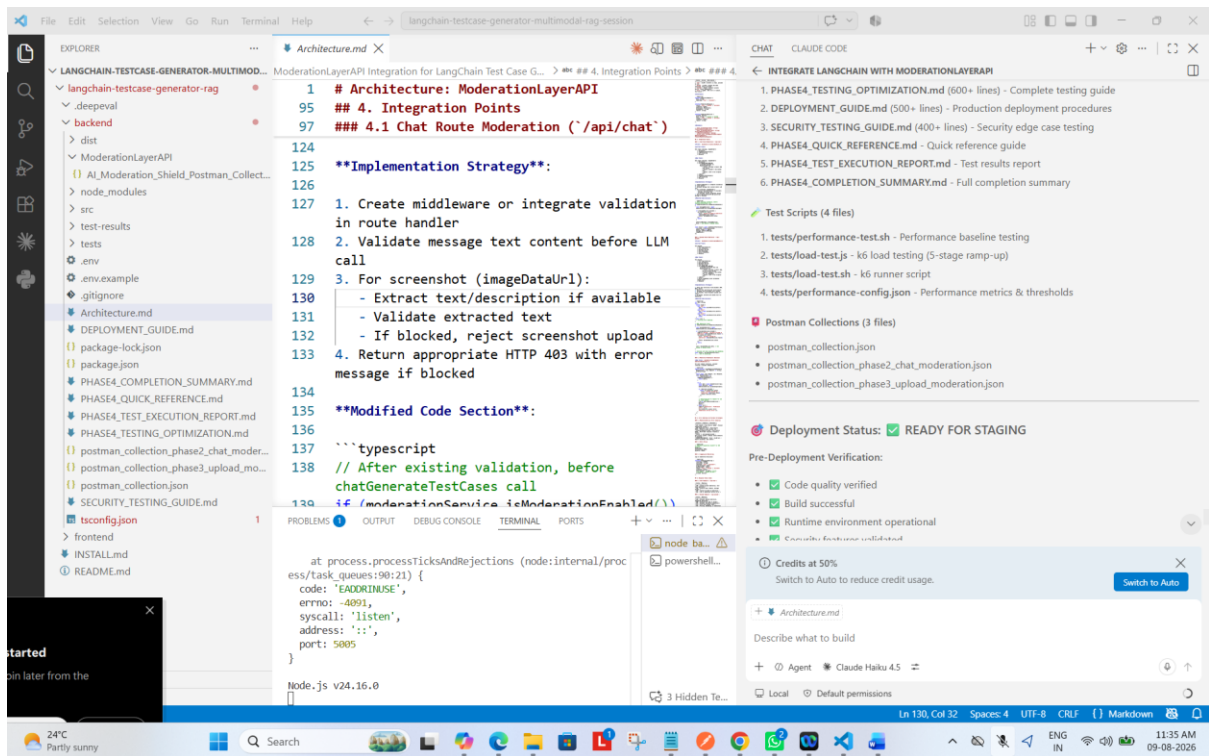
Describe what to build

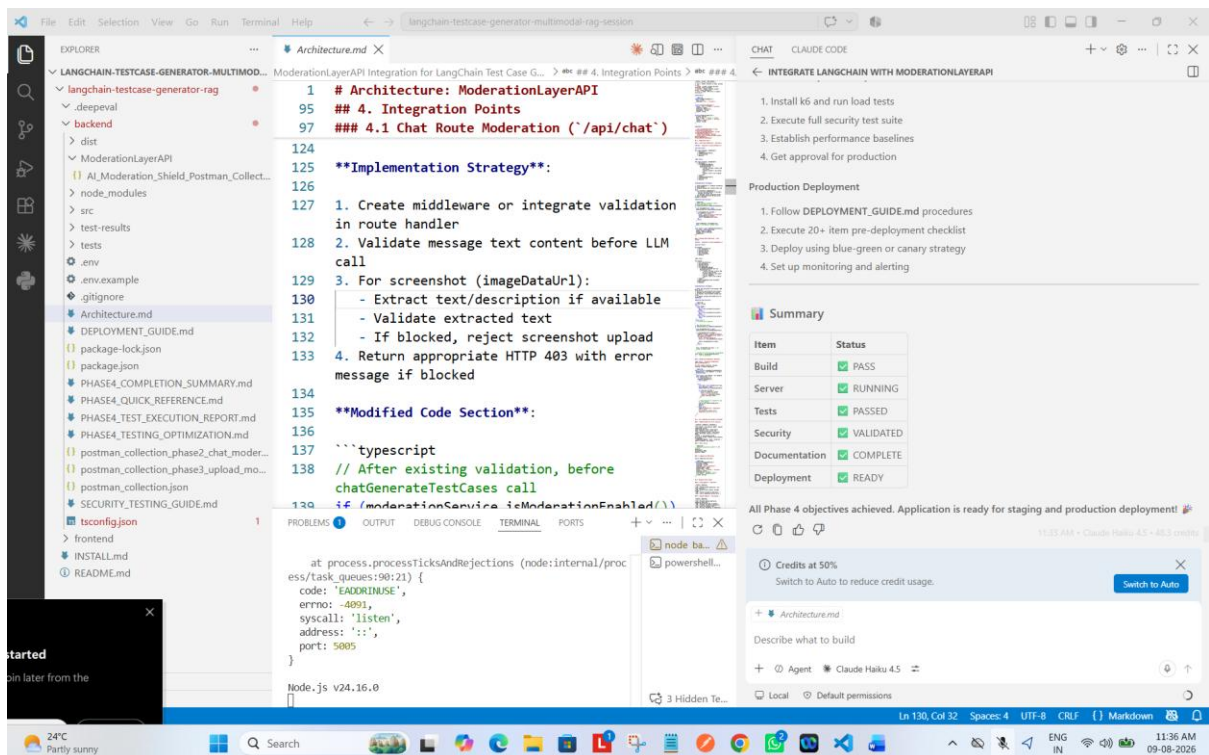
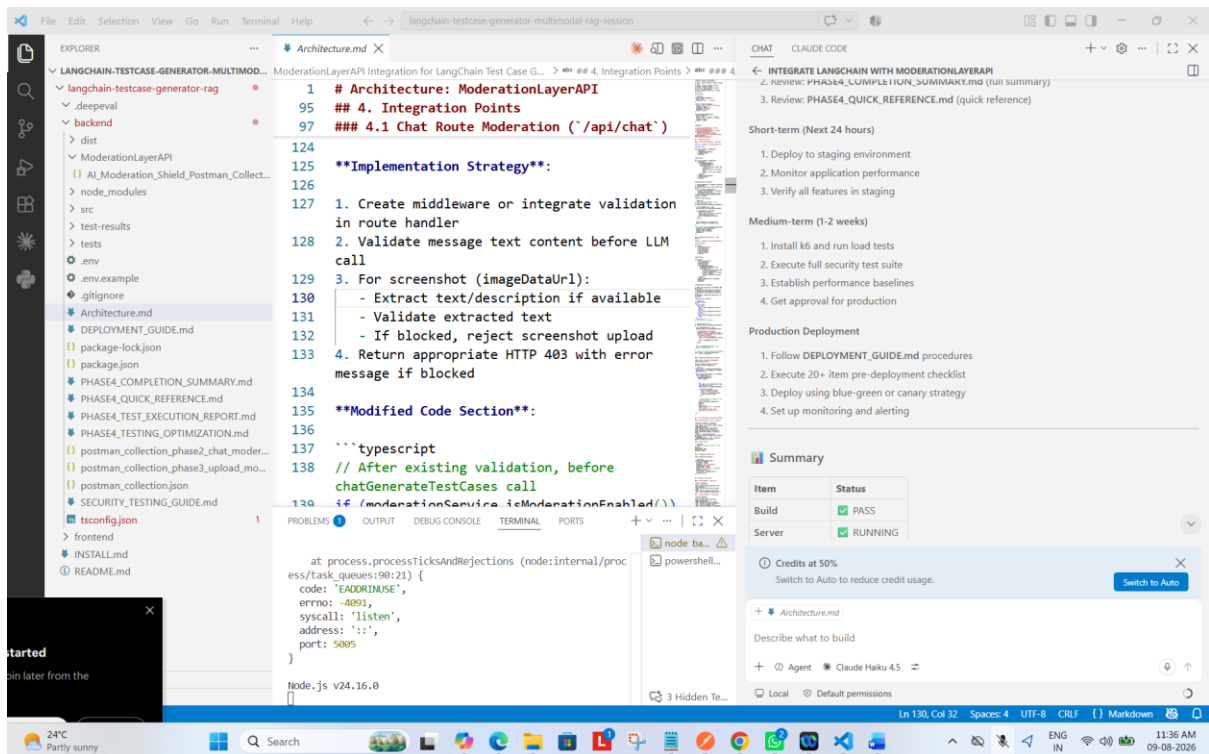
Agent Claude Haiku 4.5

Local Default permissions

Phase 4 result

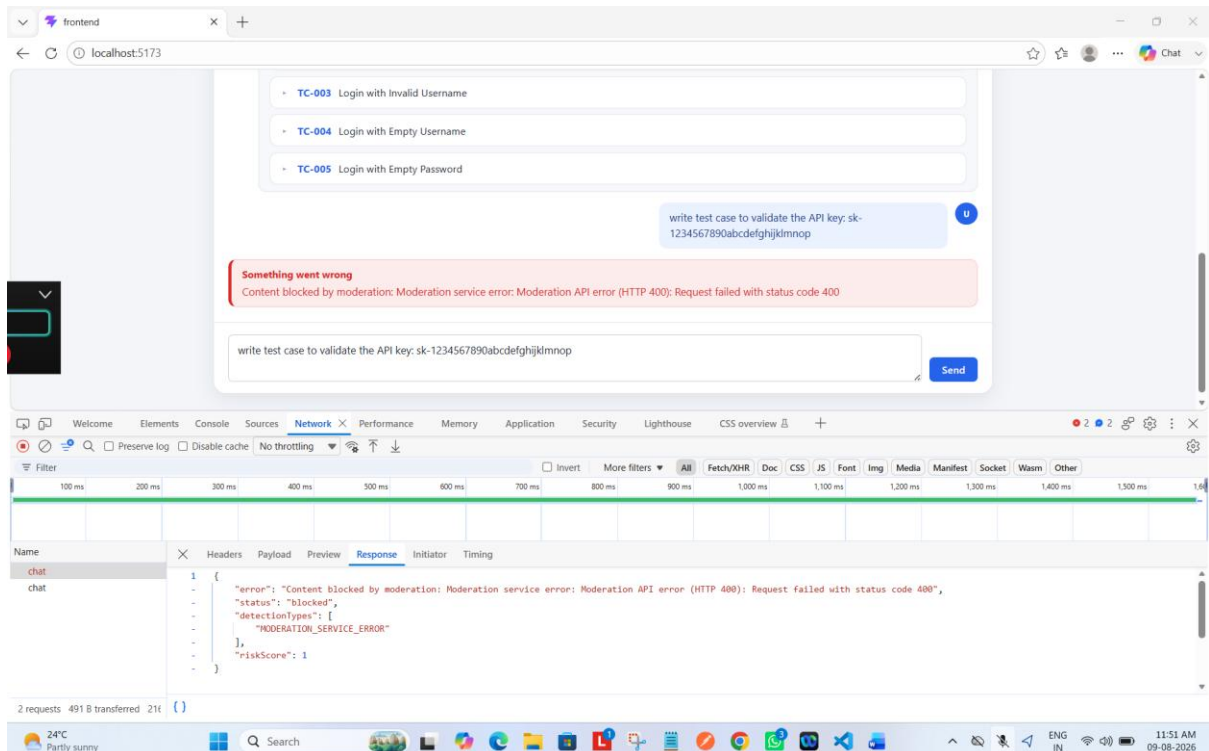
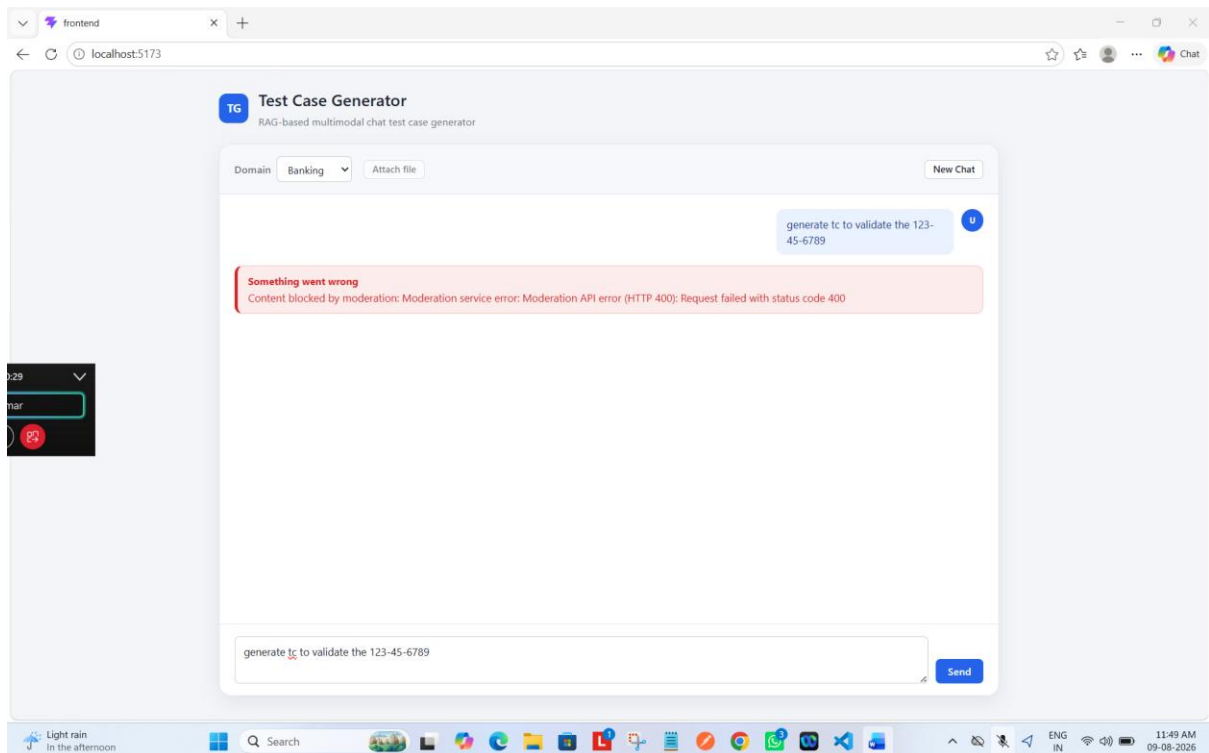






Verify the moderation layer validation in front end

SSN validation -



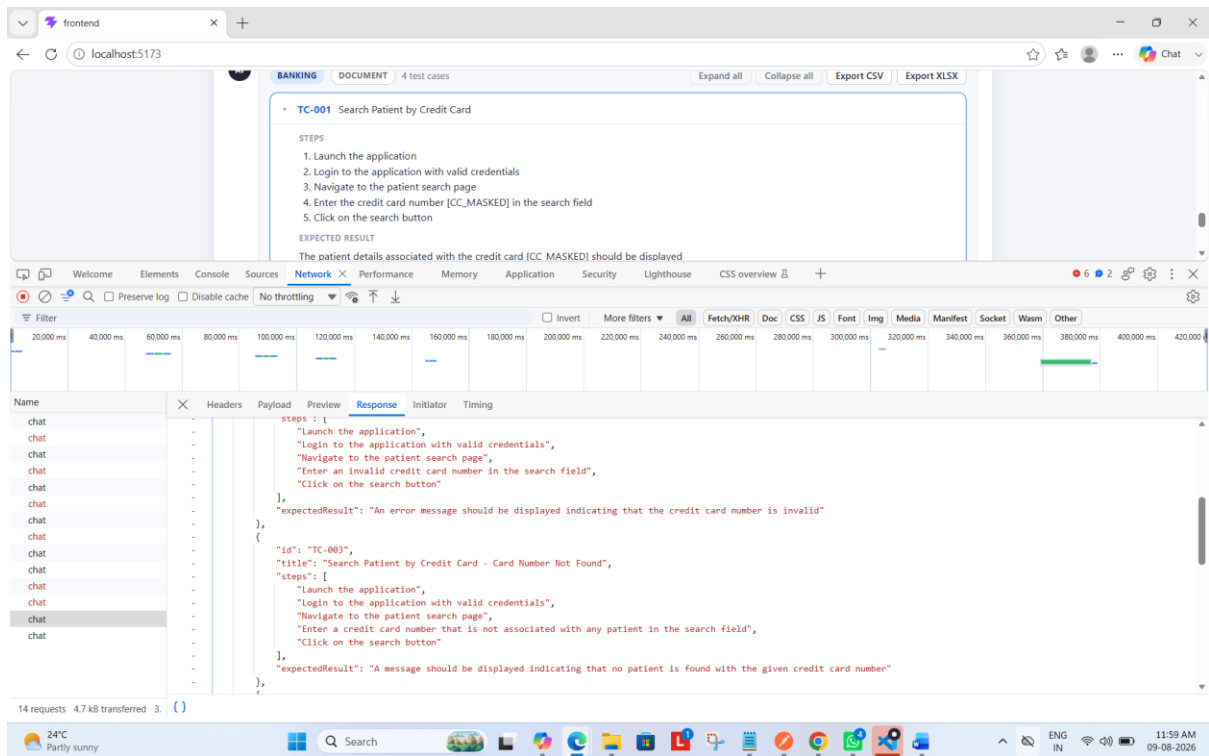
Api key - validation

The screenshot displays a web browser window with a chatbot interface. The chatbot is titled "Webex" and shows a message: "generate tc to search for the patient having credit card 4587 5689 5468 4585". Below the chatbot, there is a "BANKING" section with a "DOCUMENT" tab and "4 test cases". The test cases are:

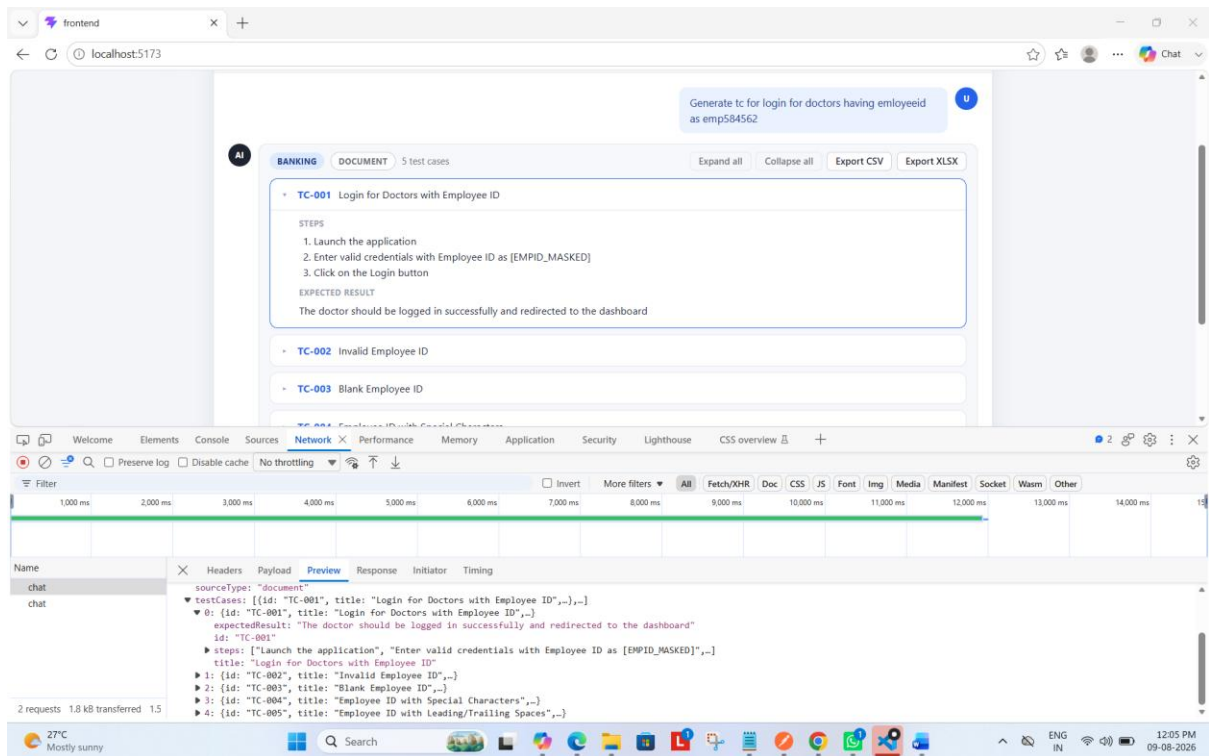
- TC-001 Search Patient by Credit Card
 - STEPS
 - 1. Launch the application
 - 2. Login to the application with valid credentials
 - 3. Navigate to the patient search page
 - 4. Enter the credit card number [CC_MASKED] in the search field
 - 5. Click on the search button
 - EXPECTED RESULT
 - The patient details associated with the credit card [CC_MASKED] should be displayed
- TC-002 Search Patient by Credit Card - Invalid Card Number
- TC-003 Search Patient by Credit Card - Card Number Not Found
- TC-004 Search Patient by Credit Card - Multiple Patients Found

Below the chatbot, there is a network monitoring tool showing a list of requests. The first request is labeled "chat" and has a status of "Success". The request URL is "http://localhost:5005/api/chat" and the request method is "OPTIONS". The response is "200 OK". The network tool also shows a timeline of requests and a list of headers.

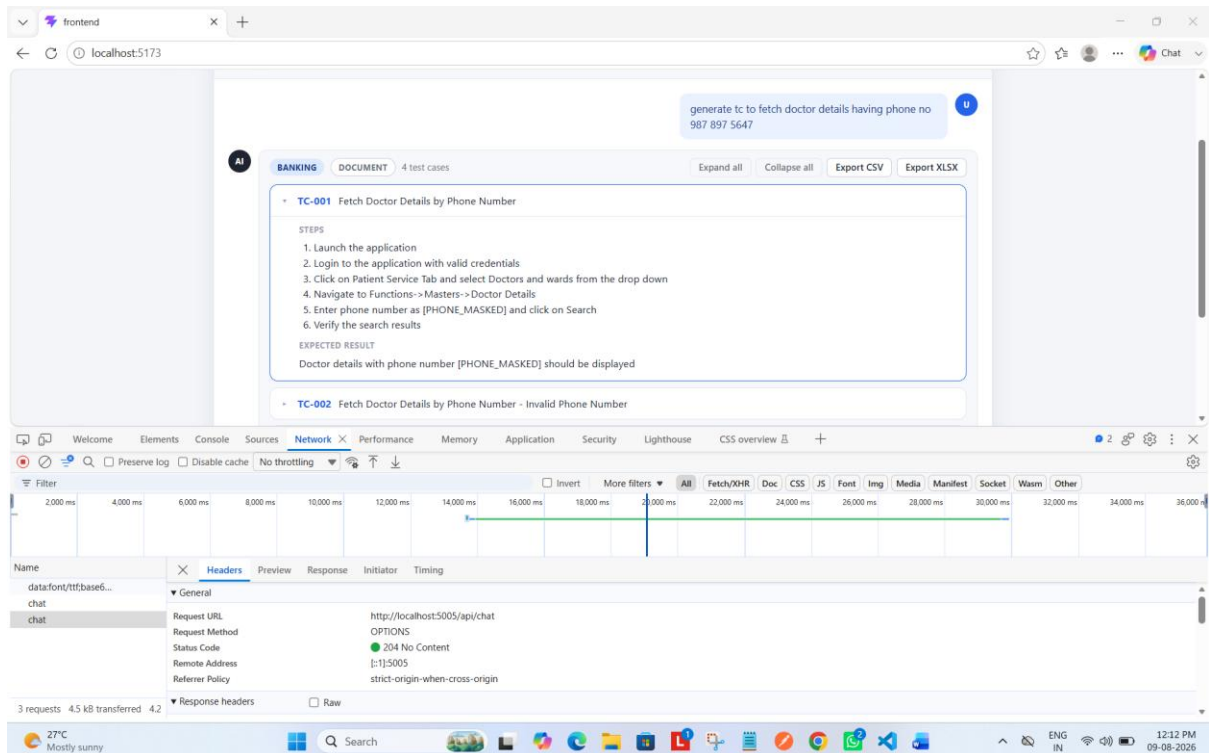
CC mask validation -



Emp id Mask validation –



Phone number mask validation –



frontend x +

localhost:5173

2. Login to the application with valid credentials
3. Click on Patient Service Tab and select Doctors and wards from the drop down
4. Navigate to Functions->Masters->Doctor Details
5. Enter phone number as [PHONE_MASKED] and click on Search
6. Verify the search results

EXPECTED RESULT
Doctor details with phone number [PHONE_MASKED] should be displayed

TC-002 Fetch Doctor Details by Phone Number - Invalid Phone Number

TC-003 Fetch Doctor Details by Phone Number - Multiple Results

TC-004 Fetch Doctor Details by Phone Number - No Results

datafont/ttfbase6...
chat
chat

Response

```
{
  "sessionId": "1a3c0ad1-83c7-4c3c-a3f0-125f0869b923",
  "domain": "banking",
  "sourceType": "document",
  "testCases": [
    {
      "id": "TC-001",
      "title": "Fetch Doctor Details by Phone Number",
      "steps": [
        "Launch the application",
        "Login to the application with valid credentials",
        "Click on Patient Service Tab and select Doctors and wards from the drop down",
        "Navigate to Functions->Masters->Doctor Details",
        "Enter phone number as [PHONE_MASKED] and click on Search",
        "Verify the search results"
      ]
    }
  ]
}
```

3 requests 4.5 kB transferred 4.2

27°C Mostly sunny

frontend x +

localhost:5173

generate tc for doctor appointment

HOSPITAL DOCUMENT 5 test cases

Expand all Collapse all Export CSV Export XLSX

TC-032 Doctor's Appointment - Schedule

datafont/ttfbase6...
chat
chat
chat
chat
chat
chat
chat

Preview

```
{
  "sessionId": "1a3c0ad1-83c7-4c3c-a3f0-125f0869b923",
  "domain": "banking",
  "sourceType": "document",
  "testCases": [
    {
      "id": "TC-001",
      "title": "Fetch Doctor Details by Phone Number",
      "steps": [
        "Launch the application",
        "Login to the application with valid credentials",
        "Click on Patient Service Tab and select Doctors and wards from the drop down",
        "Navigate to Functions->Masters->Doctor Details",
        "Enter phone number as [PHONE_MASKED] and click on Search",
        "Verify the search results"
      ]
    }
  ]
}
```

9 requests 12.5 kB transferred 1

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Email validation –

frontend x +

localhost:5173

generate tc to fetch doctor having mail id abcd@gmail.com

HOSPITAL DOCUMENT 4 test cases Expand all Collapse all Export CSV Export XLSX

TC-001 Fetch Doctor by Email ID

STEPS

1. Launch the application
2. Login to the application with valid credentials
3. Click on Patient Service Tab and select Doctors and wards from the drop down
4. Navigate to Functions->Transactions->Doctor Dashboard
5. Click on Search Icon
6. Enter email ID [EMAIL_MASKED] and click on Search
7. Verify the search results

EXPECTED RESULT

The doctor details should be displayed with the matching email ID [EMAIL_MASKED]

TC-002 Fetch Doctor by Email ID - Invalid Email

20,000 ms 40,000 ms 60,000 ms 80,000 ms 100,000 ms 120,000 ms 140,000 ms 160,000 ms 180,000 ms 200,000 ms 220,000 ms 240,000 ms 260,000 ms

Name X Headers Payload Preview Response Initiator Timing

1 {

```
"sessionId": "5e1b2a2a-3c33-4378-b3cd-0f9422ddff8",
"domain": "hospital",
"sourceType": "document",
"testCases": [
  {
    "id": "TC-001",
    "title": "Fetch Doctor by Email ID",
    "steps": [
```

7 requests 9.1 kB transferred 8.3

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Search

ENG IN 12:15 PM 09-08-2026

frontend x +

localhost:5173

generate tc to fetch doctor having mail id abcd@gmail.com

HOSPITAL DOCUMENT 4 test cases Expand all Collapse all Export CSV Export XLSX

TC-001 Fetch Doctor by Email ID

STEPS

1. Launch the application
2. Login to the application with valid credentials
3. Click on Patient Service Tab and select Doctors and wards from the drop down
4. Navigate to Functions->Transactions->Doctor Dashboard
5. Click on Search Icon
6. Enter email ID [EMAIL_MASKED] and click on Search
7. Verify the search results

EXPECTED RESULT

The doctor details should be displayed with the matching email ID [EMAIL_MASKED]

TC-002 Fetch Doctor by Email ID - Invalid Email

20,000 ms 40,000 ms 60,000 ms 80,000 ms 100,000 ms 120,000 ms 140,000 ms 160,000 ms 180,000 ms 200,000 ms 220,000 ms 240,000 ms 260,000 ms

Name X Headers Payload Preview Response Initiator Timing

1 {

```
"Launch the application",
"Login to the application with valid credentials",
"Click on Patient Service Tab and select Doctors and wards from the drop down",
"Navigate to Functions->Transactions->Doctor Dashboard",
"Click on Search Icon",
"Enter email ID [EMAIL_MASKED] and click on Search",
"Verify the search results"
},
"expectedResult": "The doctor details should be displayed with the matching email ID [EMAIL_MASKED]"
```

7 requests 9.1 kB transferred 8.3

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Search

ENG IN 12:15 PM 09-08-2026

frontend x +

localhost:5173

generate tc for doctor appointment

AI HOSPITAL DOCUMENT 5 test cases Expand all Collapse all Export CSV Export XLSX

- TC_032 Doctor's Appointment - Schedule
- TC_033 Doctor's Appointment - Walkin
- TC_2835 Transactions - Dashboard - Dashboard tab - Registration - Doctor Appointment-Schedule
- TC_1123 Consult an patient for which appointment is not available
- TC_2515 Secretary Dashboard - Appointment with Doctor - Schedule

Describe requirements or ask for refinements...

Network Performance Memory Application Security Lighthouse CSS overview

Filter 20,000 ms 40,000 ms 60,000 ms 80,000 ms 100,000 ms 120,000 ms 140,000 ms 160,000 ms 180,000 ms 200,000 ms 220,000 ms 240,000 ms 260,000 ms 280,000 ms 300,000 ms 320,000 ms 340,000 ms

Name chat chat chat chat chat chat chat

9 requests 12.5 kB transferred 1

```
{
  "sessionId": "5e1b2a2a-3c33-4378-b3cd-0fb9422ddf8",
  "domain": "hospital",
  "sourceType": "document",
  "testCases": [
    {
      "id": "TC-001",
      "title": "Fetch Doctor by Email ID",
      "description": "Fetch Doctor by Email ID - Invalid Email",
      "steps": [
        {
          "id": "TC-001",
          "title": "Fetch Doctor by Email ID - Invalid Email",
          "description": "Fetch Doctor by Email ID - Invalid Email"
        }
      ]
    },
    {
      "id": "TC-002",
      "title": "Fetch Doctor by Email ID - Empty Search",
      "description": "Fetch Doctor by Email ID - Empty Search",
      "steps": [
        {
          "id": "TC-002",
          "title": "Fetch Doctor by Email ID - Empty Search",
          "description": "Fetch Doctor by Email ID - Empty Search"
        }
      ]
    },
    {
      "id": "TC-003",
      "title": "Fetch Doctor by Email ID - Multiple Matches",
      "description": "Fetch Doctor by Email ID - Multiple Matches",
      "steps": [
        {
          "id": "TC-003",
          "title": "Fetch Doctor by Email ID - Multiple Matches",
          "description": "Fetch Doctor by Email ID - Multiple Matches"
        }
      ]
    }
  ]
}
```

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Search

ENG IN 12:18 PM 09-08-2026

Safe content – No moderation

frontend x +

localhost:5173

generate tc for doctor appointment

AI HOSPITAL DOCUMENT 5 test cases Expand all Collapse all Export CSV Export XLSX

- TC_032 Doctor's Appointment - Schedule
- TC_033 Doctor's Appointment - Walkin
- TC_2835 Transactions - Dashboard - Dashboard tab - Registration - Doctor Appointment-Schedule
- TC_1123 Consult an patient for which appointment is not available
- TC_2515 Secretary Dashboard - Appointment with Doctor - Schedule

Describe requirements or ask for refinements...

Network Performance Memory Application Security Lighthouse CSS overview

Filter 20,000 ms 40,000 ms 60,000 ms 80,000 ms 100,000 ms 120,000 ms 140,000 ms 160,000 ms 180,000 ms 200,000 ms 220,000 ms 240,000 ms 260,000 ms 280,000 ms 300,000 ms 320,000 ms 340,000 ms

Name chat chat chat chat chat chat chat

9 requests 12.5 kB transferred 1

```
{
  "sessionId": "5e1b2a2a-3c33-4378-b3cd-0fb9422ddf8",
  "domain": "hospital",
  "sourceType": "document",
  "testCases": [
    {
      "id": "TC-032",
      "title": "Doctor's Appointment - Schedule",
      "description": "Doctor's Appointment - Schedule",
      "steps": [
        {
          "id": "TC-032",
          "title": "Doctor's Appointment - Schedule",
          "description": "Doctor's Appointment - Schedule"
        }
      ]
    },
    {
      "id": "TC-033",
      "title": "Doctor's Appointment - Walkin",
      "description": "Doctor's Appointment - Walkin",
      "steps": [
        {
          "id": "TC-033",
          "title": "Doctor's Appointment - Walkin",
          "description": "Doctor's Appointment - Walkin"
        }
      ]
    },
    {
      "id": "TC-2835",
      "title": "Transactions - Dashboard - Dashboard tab - Registration - Doctor Appointment-Schedule",
      "description": "Transactions - Dashboard - Dashboard tab - Registration - Doctor Appointment-Schedule",
      "steps": [
        {
          "id": "TC-2835",
          "title": "Transactions - Dashboard - Dashboard tab - Registration - Doctor Appointment-Schedule",
          "description": "Transactions - Dashboard - Dashboard tab - Registration - Doctor Appointment-Schedule"
        }
      ]
    },
    {
      "id": "TC-1123",
      "title": "Consult an patient for which appointment is not available",
      "description": "Consult an patient for which appointment is not available",
      "steps": [
        {
          "id": "TC-1123",
          "title": "Consult an patient for which appointment is not available",
          "description": "Consult an patient for which appointment is not available"
        }
      ]
    },
    {
      "id": "TC-2515",
      "title": "Secretary Dashboard - Appointment with Doctor - Schedule",
      "description": "Secretary Dashboard - Appointment with Doctor - Schedule",
      "steps": [
        {
          "id": "TC-2515",
          "title": "Secretary Dashboard - Appointment with Doctor - Schedule",
          "description": "Secretary Dashboard - Appointment with Doctor - Schedule"
        }
      ]
    }
  ]
}
```

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Search

ENG IN 12:17 PM 09-08-2026

